Deep Learning 1

EECS 492

Introduction to Artificial Intelligence

Credits: Many slides from Prof. Jia-Bin Huang, Prof. Fei Fei Li, Prof. Christopher Manning and Prof. Dan Jurafsky

Reminders

- HW1 due tomorrow!
- This week's topic for discussion section: deep learning

Reminders

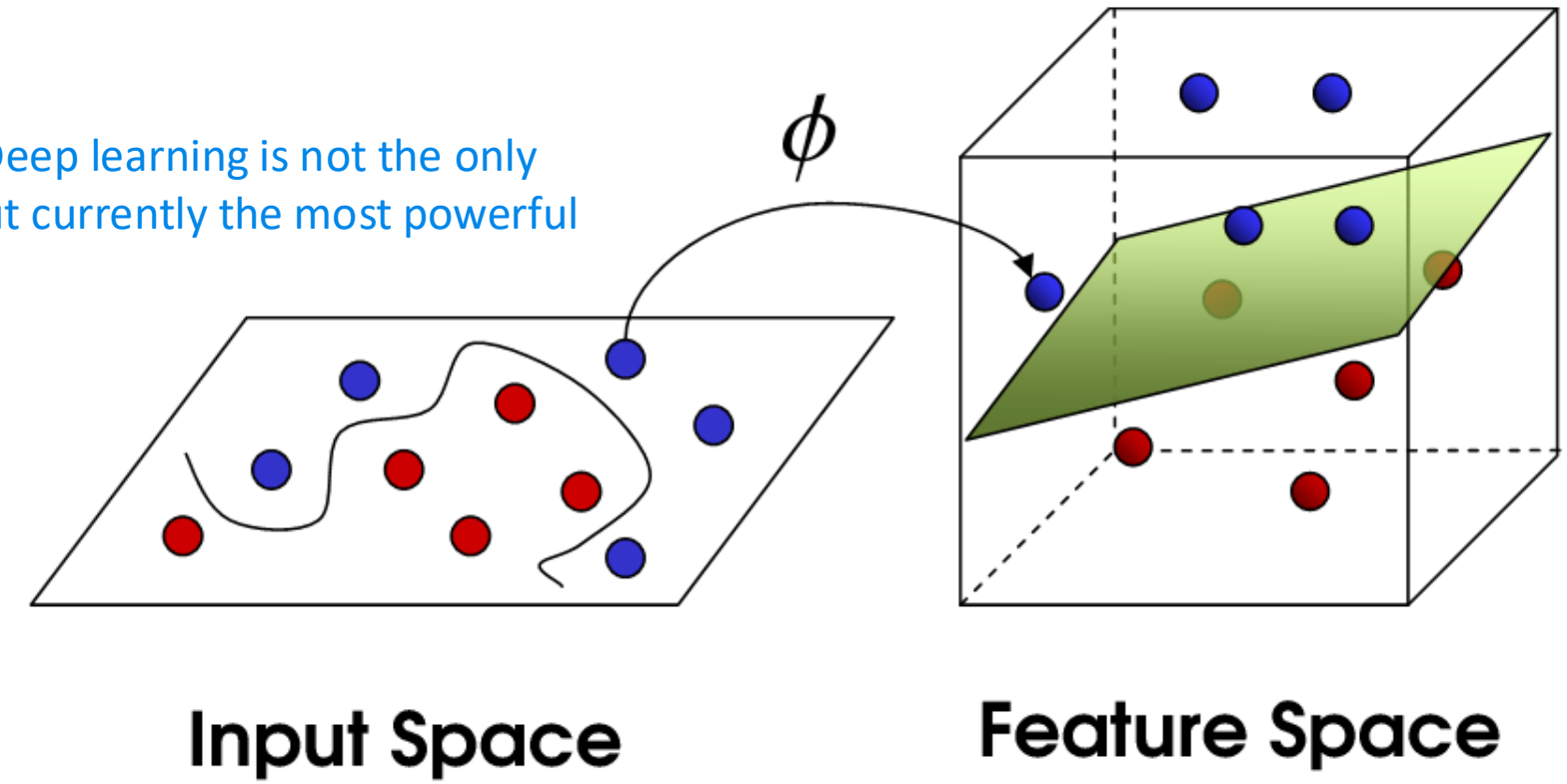
- Ethics discussion on February 13
  - Complete reading quiz before the session
  - Make-up: come to professor office hours the following week (come at the beginning of office hours)
    - 1-1:30 PM on M-W-TH

Today's Class

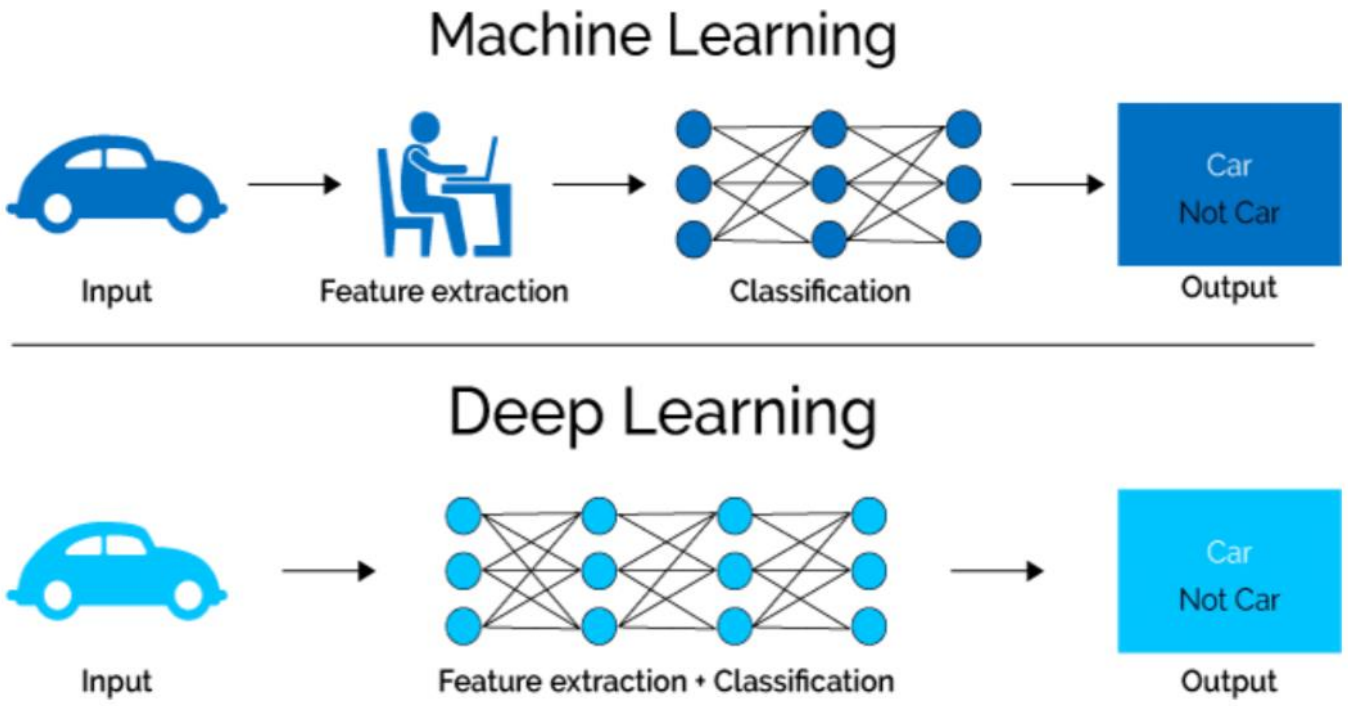
- Neural Networks
- Convolutional Neural Nets (CNNs)

Neural Networks Enable Automated Feature Engineering

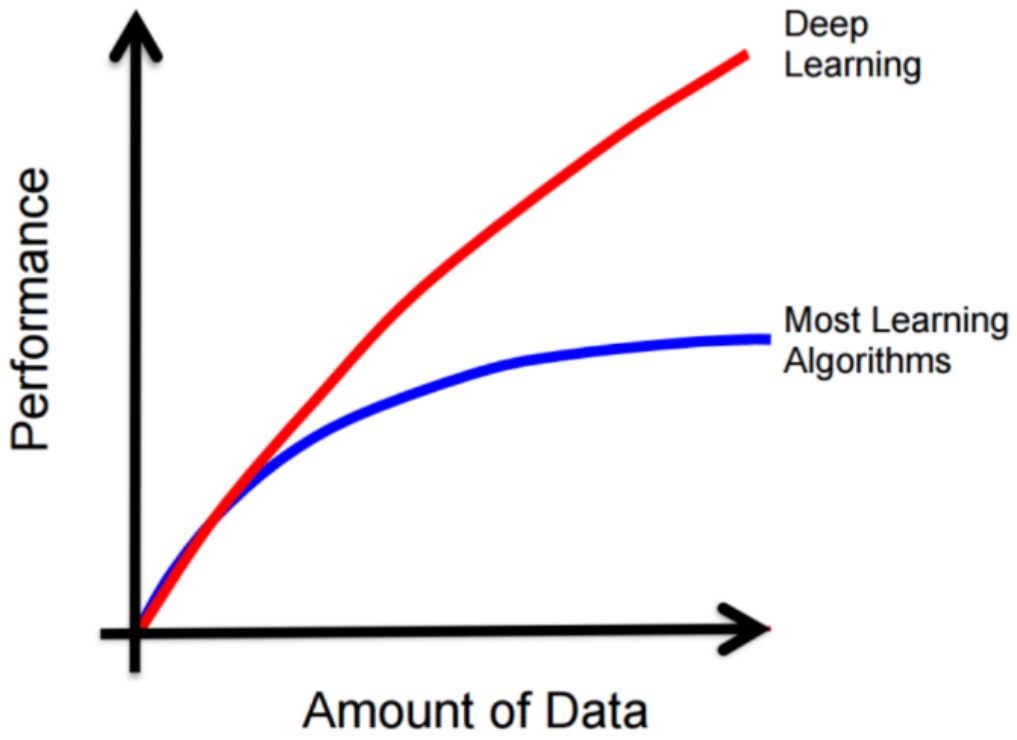
Note: Deep learning is not the only one, but currently the most powerful



Deep Learning Automates the Creation of Knowledge Representations



https://deeplearning.mit.edu/



Neural Network Unit

- Take weighted sum of inputs, plus a bias

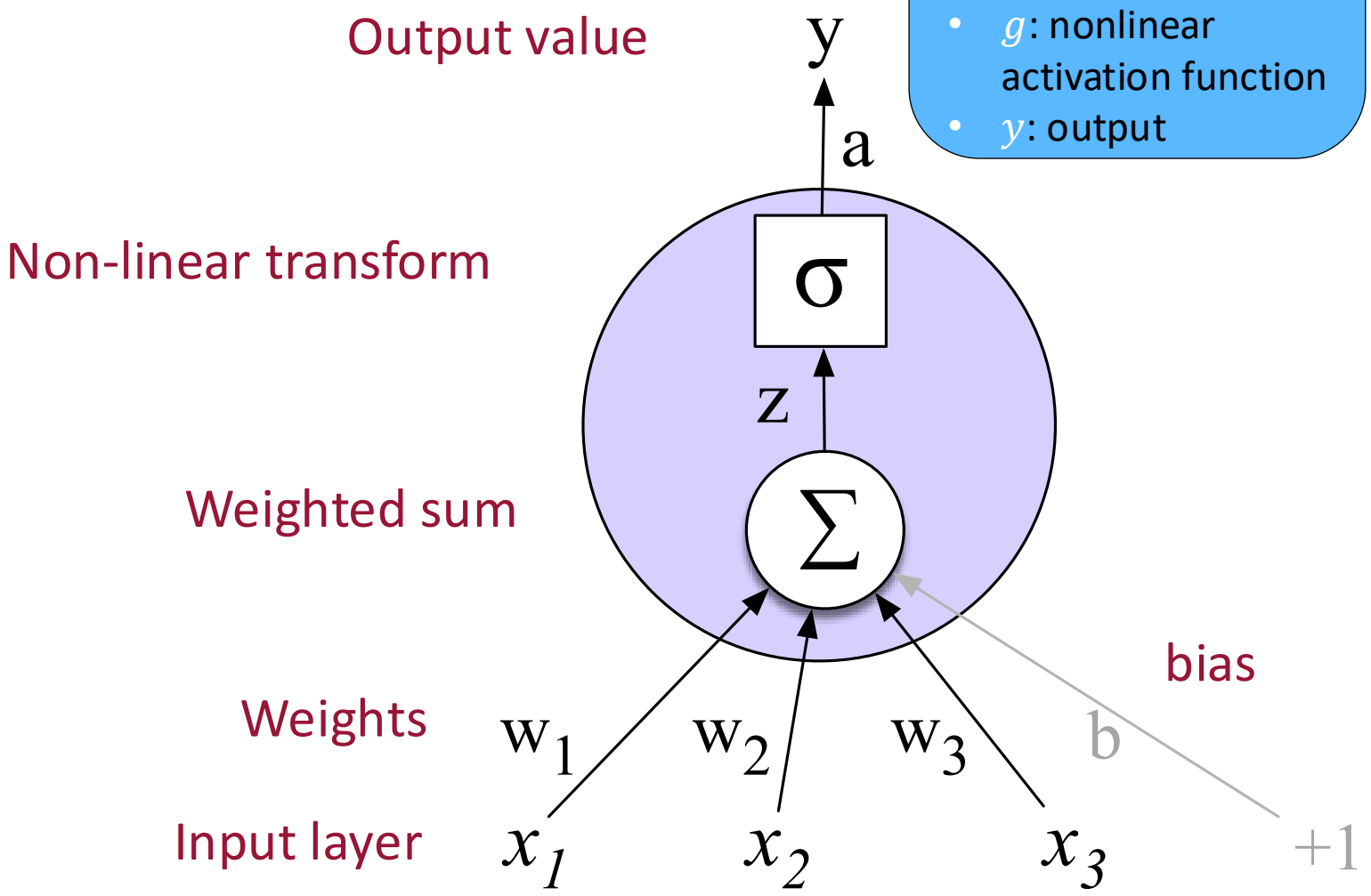
$$z = b + \sum_i w_i x_i$$

$$z = w \cdot x + b$$

- Instead of just using z, we'll apply a nonlinear activation function g:

$$y = a = g(z)$$

- x: input vector
- w: weight vector
- b: bias term
- z: output before nonlinear function
- g: nonlinear activation function
- y: output

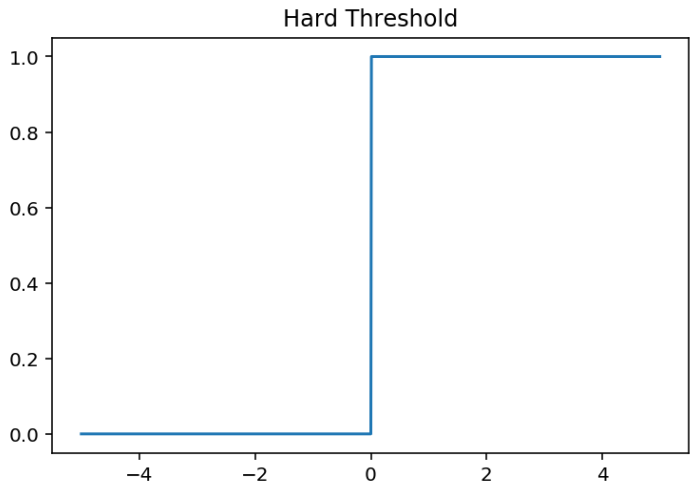


Types of Activation Functions

- Hard threshold

$$g(s) = 1 \text{ iff } s > 0, \text{else } 0$$

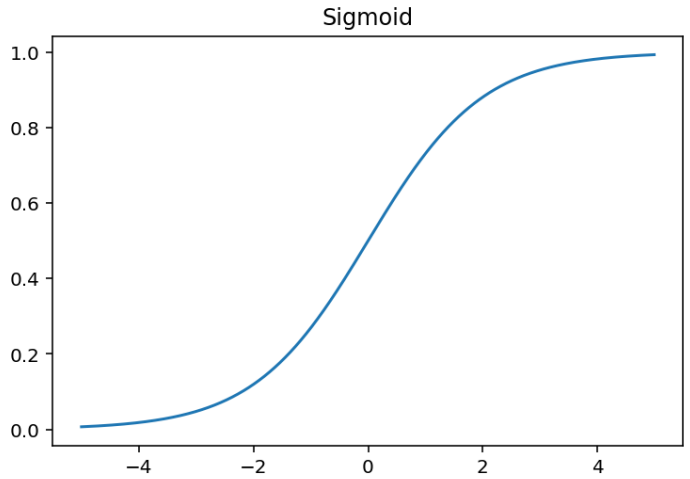
Non-differentiable!



- Sigmoid threshold

$$g(s) = \frac{1}{1 + e^{-s}}$$

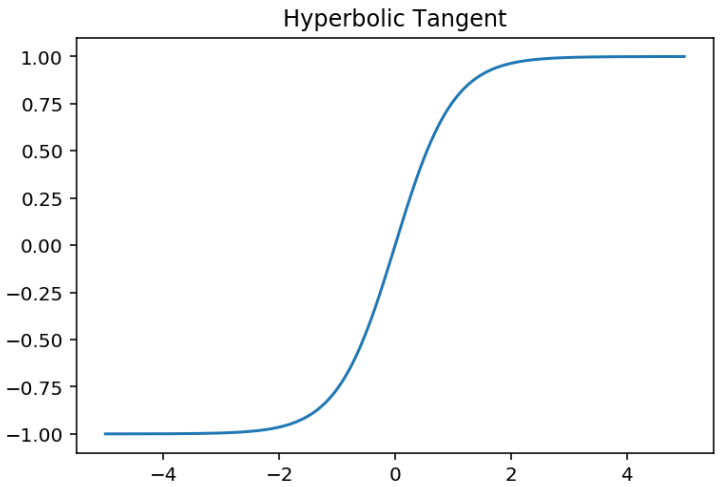
$$g'(s) = g(s)(1 - g(s))$$



Types of Activation Functions

- Hyperbolic Tangent (tanh)

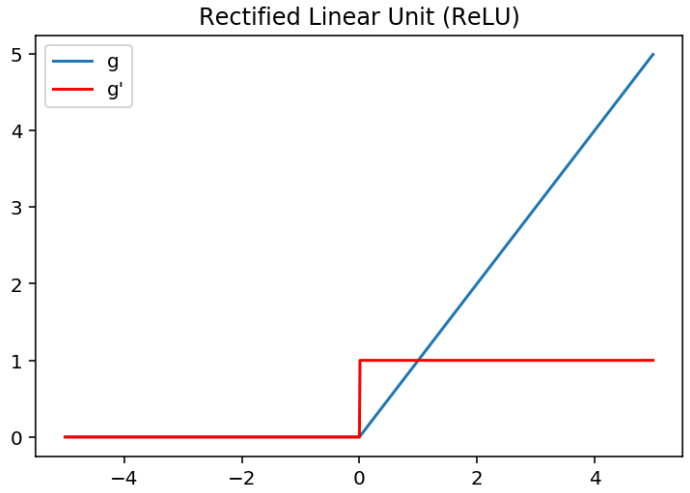
$$g(s) = \frac{e^{s} - e^{-s}}{e^{s} + e^{-s}}$$



- Rectified Linear Unit (ReLU)

$$g(s) = \max(0, s)$$

$$g'(s) = \begin{cases} 1, & s > 0 \\ 0, & otherwise \end{cases}$$



This is just the hard threshold.
Leads to sparse activations!

When poll is active respond at **PollEv.com/492**

Example

- Suppose a unit has:
  - $\mathbf{w} = [0.2, 0.3, 0.9]$
  - $b = 0.5$
  - Sigmoid activation function

$$\sigma(s) = \frac{1}{1 + e^{-s}}$$

- What is the output y for input vector $\mathbf{x} = [0.5, 0.6, 0.1]$?

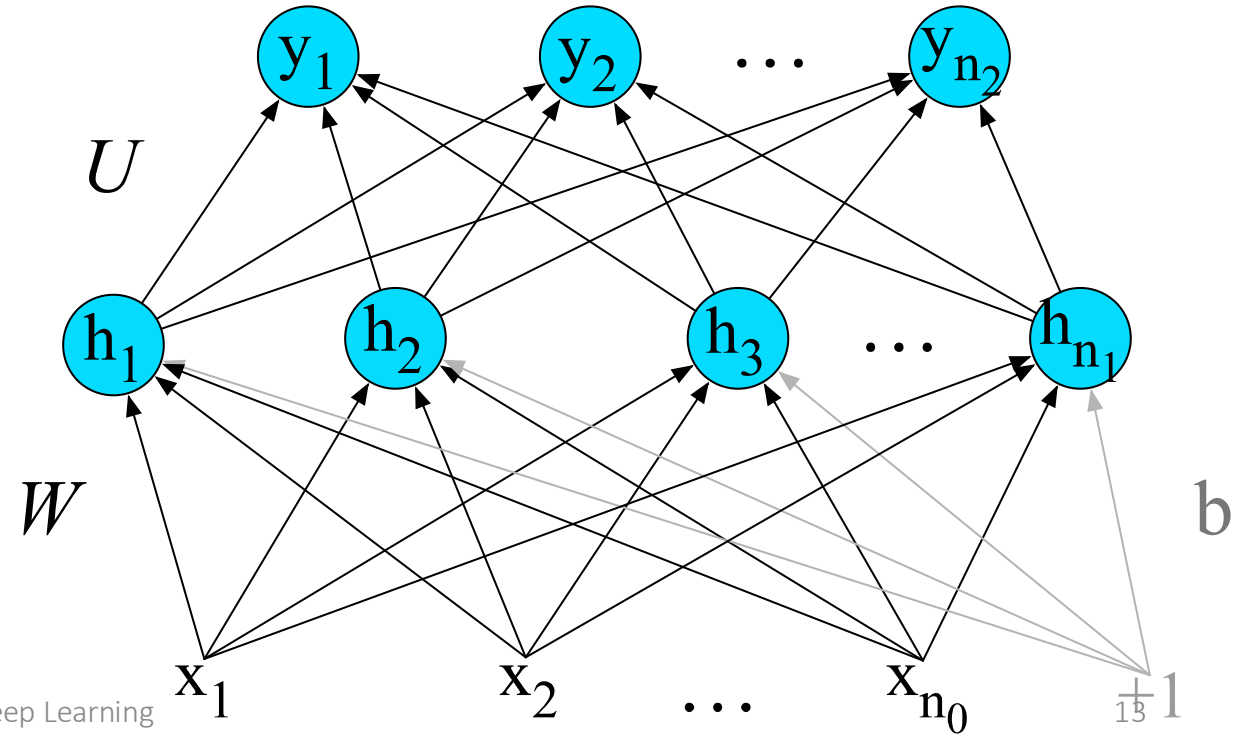
- x_i: ith input
- W, U: weight matrices

Feedforward Neural Networks (FNN)

- Multilayer network in which the units are connected with no cycles
  - Outputs from units in each layer are passed to units in the next higher layer
  - No outputs are passed back to lower layers

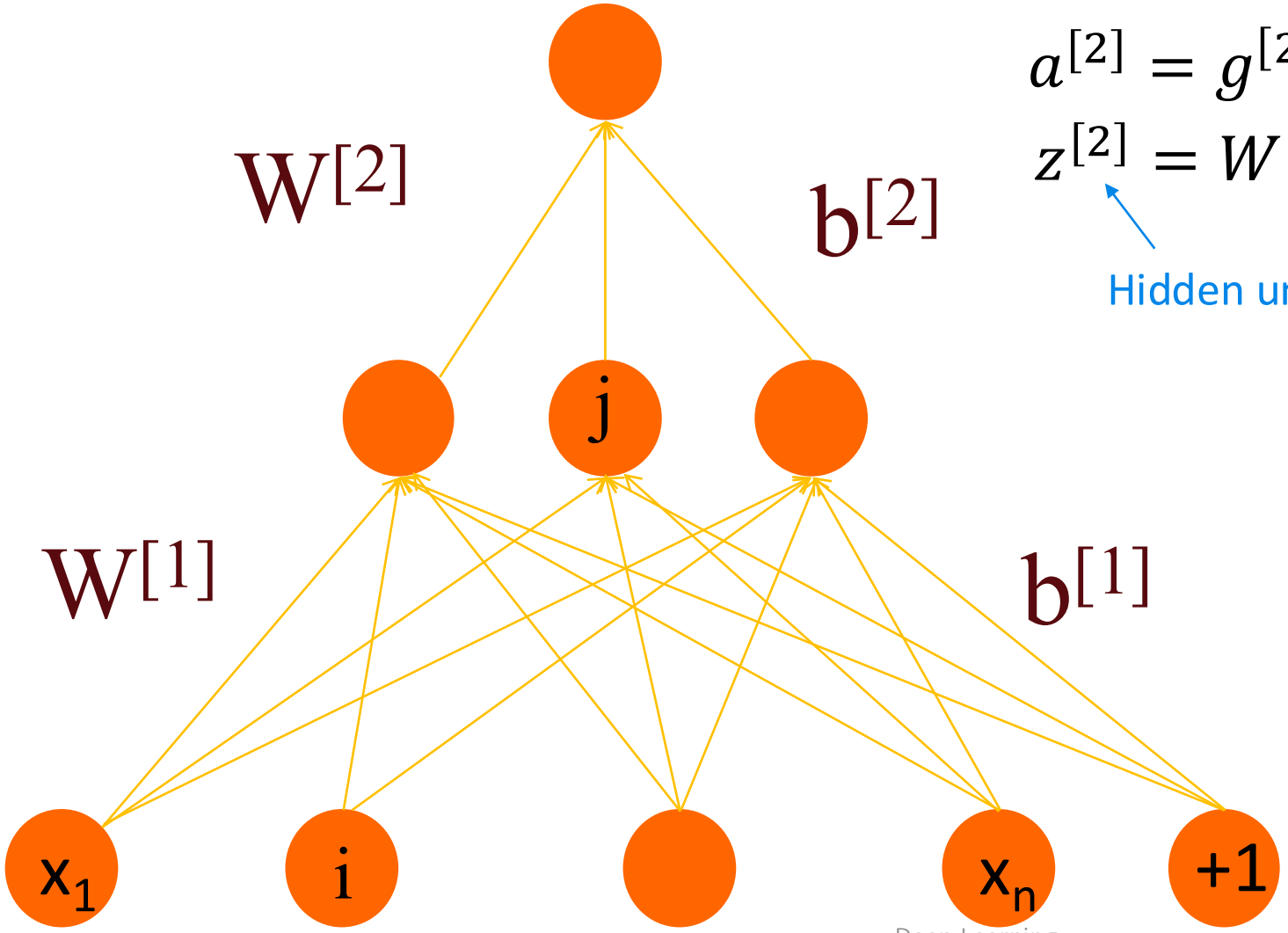
- Each layer is fully-connected
  - Link between every pair of units from 2 adjacent layers

- Also known as:
  - Multi-layer perceptron (**MLP**)
  - Fully-connected Neural Net (**FCNN**)



- x_i: ith input
- W, U: weight matrices

Multi-layer Notation



$$y = a^{[2]}$$

$$a^{[2]} = g^{[2]}(z^{[2]})$$

$$z^{[2]} = W^{[2]}a^{[1]} + b^{[2]}$$

Hidden unit, often denoted with letter z or h

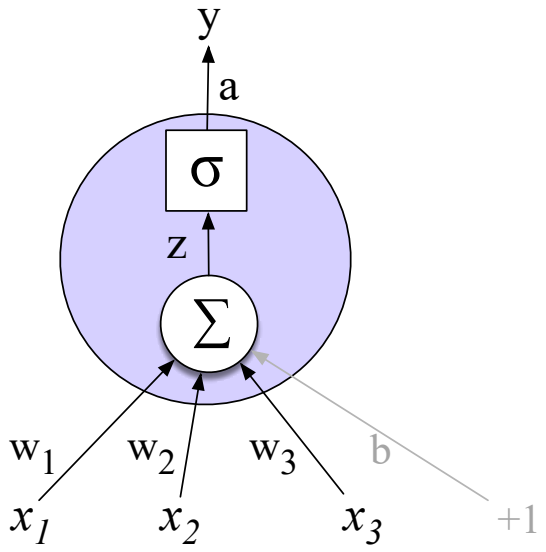
$$a^{[1]} = g^{[1]}(z^{[1]})$$

$$z^{[1]} = W^{[1]}a^{[0]} + b^{[1]}$$

$a^{[0]}$

Multi Layer Notation

$$
\begin{aligned}
z^{[1]} &= W^{[1]}a^{[0]} + b^{[1]} \\
a^{[1]} &= g^{[1]}(z^{[1]}) \\
z^{[2]} &= W^{[2]}a^{[1]} + b^{[2]} \\
a^{[2]} &= g^{[2]}(z^{[2]}) \\
\hat{y} &= a^{[2]}
\end{aligned}
$$



for i **in** 1..n

$$
\begin{aligned}
z^{[i]} &= W^{[i]}\, a^{[i-1]} + b^{[i]} \\
a^{[i]} &= g^{[i]}(z^{[i]}) \\
\hat{y} &= a^{[n]}
\end{aligned}
$$

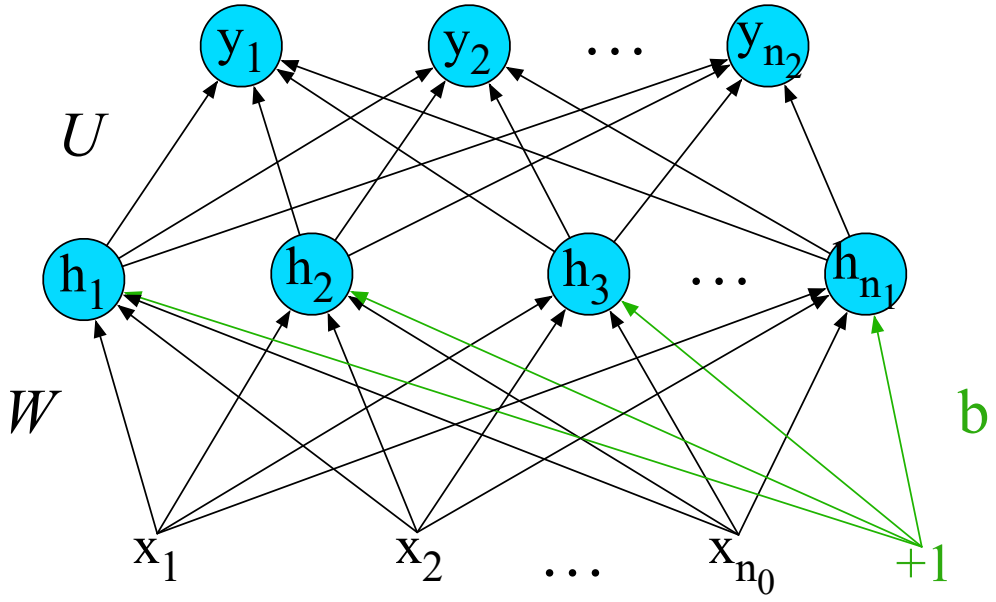
Replacing the bias unit

- Let's switch to a notation without the bias unit
- Just a notational change

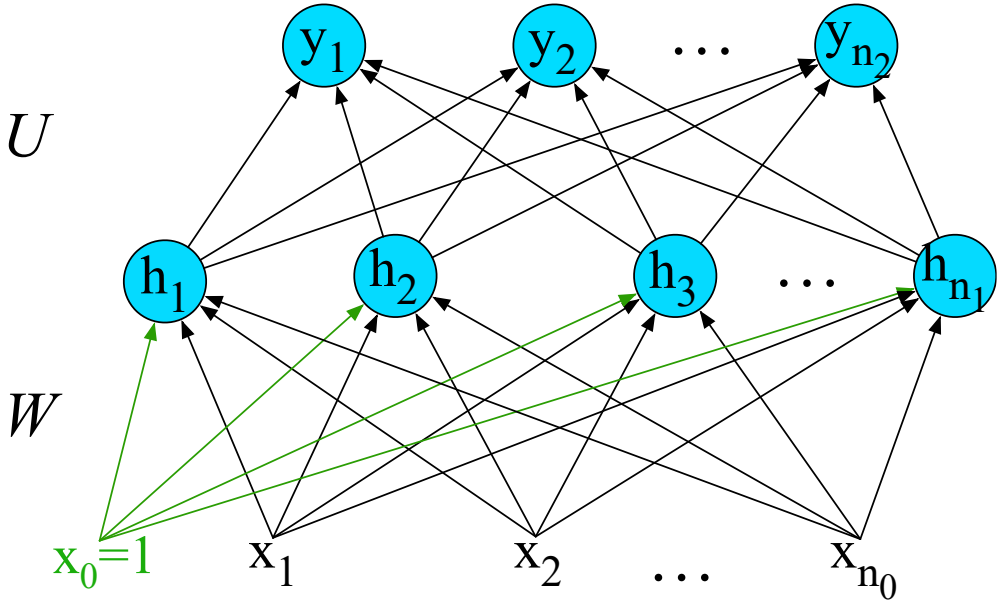
1. Add a dummy node $a_0=1$ to each layer
2. Its weight w_0 will be the bias
3. So input layer $a^{[0]}_0=1$,
   - And $a^{[1]}_0=1$, $a^{[2]}_0=1$,…

Replacing the bias unit

Instead of:



We'll do this:



Replacing the bias unit

- Instead of:

$$x = x_1, x_2, \ldots, x_{n0}$$

$$h = g\left(\sum_{i=1}^{n_0} W_{ji} x_i + b_j\right)$$

$$h = g(Wx + b)$$

We'll do this:

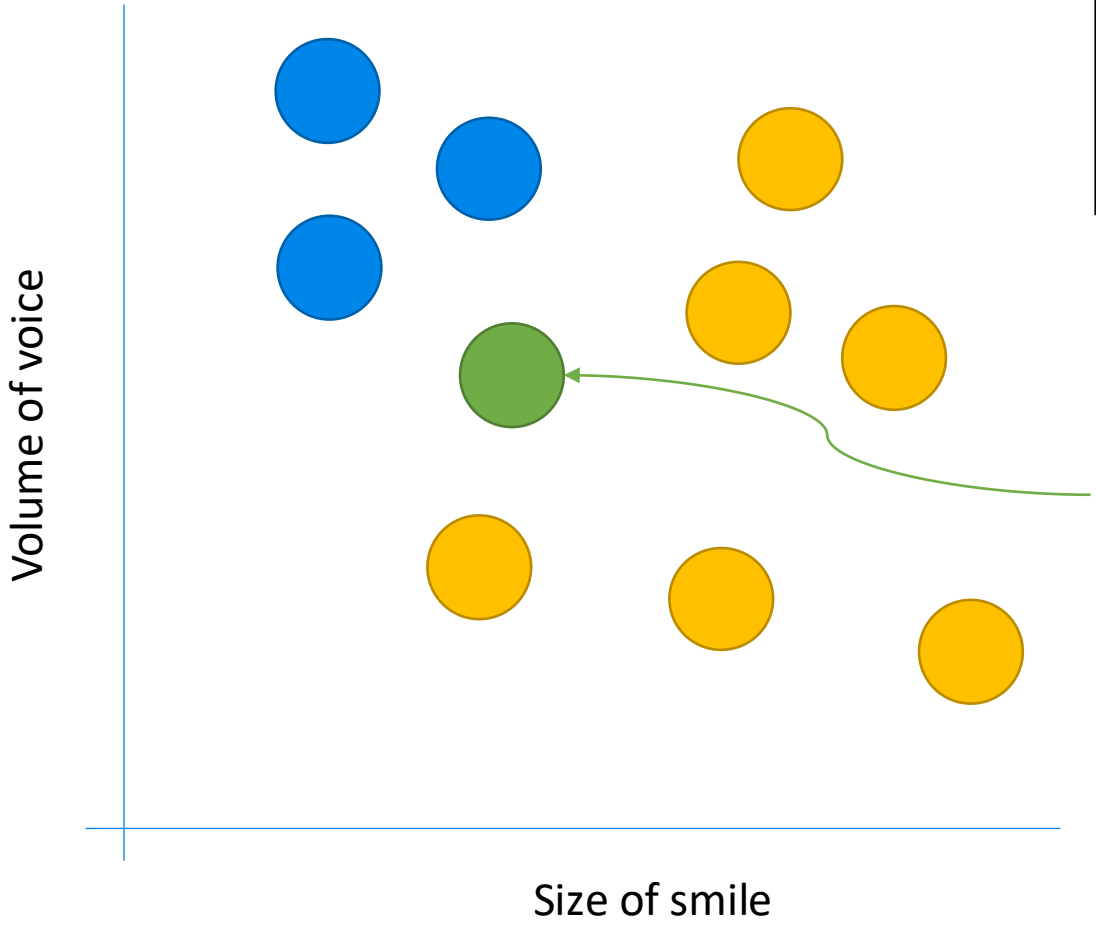
$$x = \boxed{x_0}, x_1, x_2, \ldots, x_{n0}$$

$$h = g\left(\sum_{i=0}^{n_0} W_{ji} x_i + b_j\right)$$

$$h = g(Wx)$$

Easier to handle!
(do this only for the input layer)

A toy example: Am I happy?



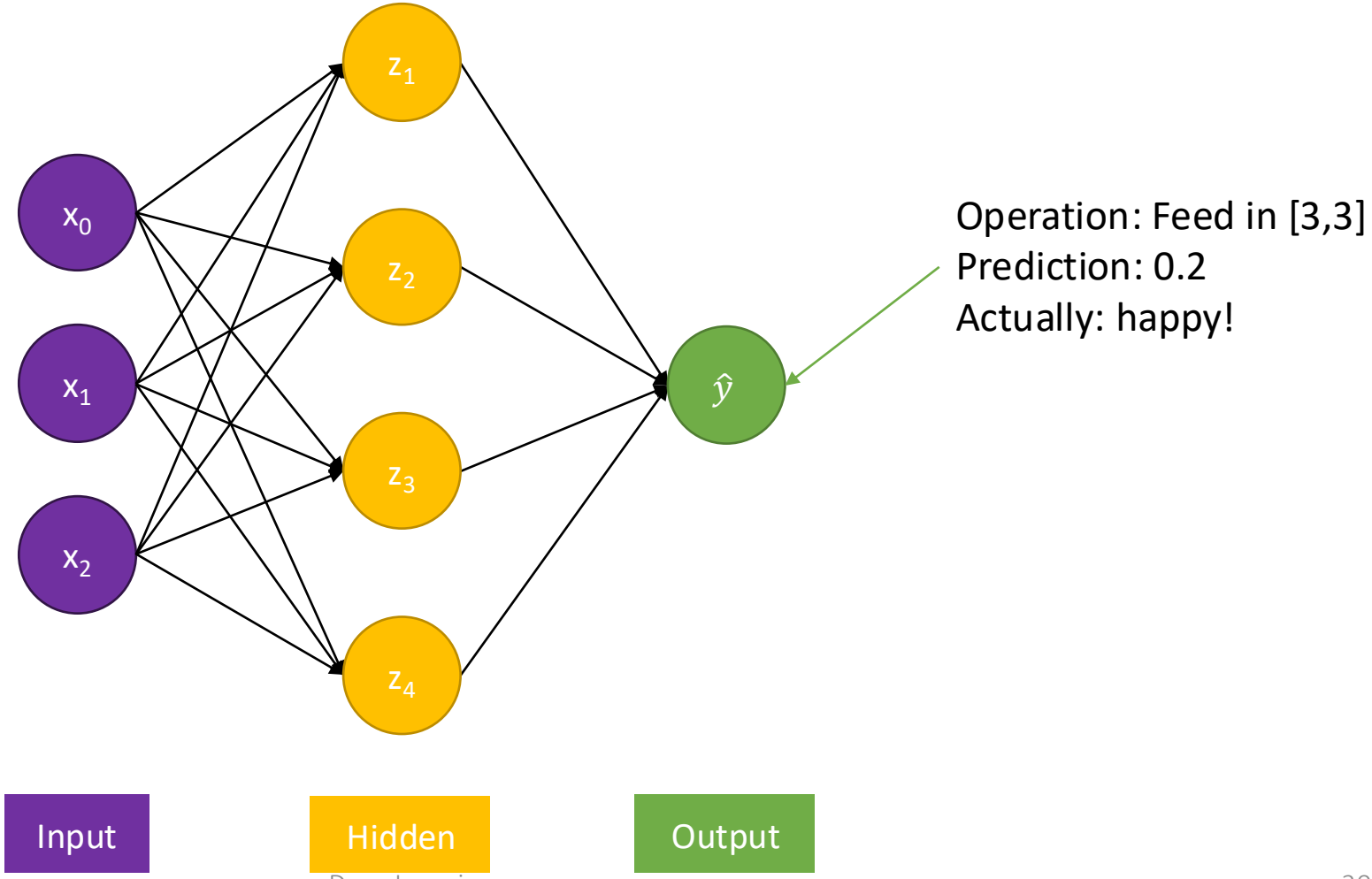
Legend:

- Happy
- Not happy

Variable change:
Let **Happy** be 1 and **Not happy** be 0

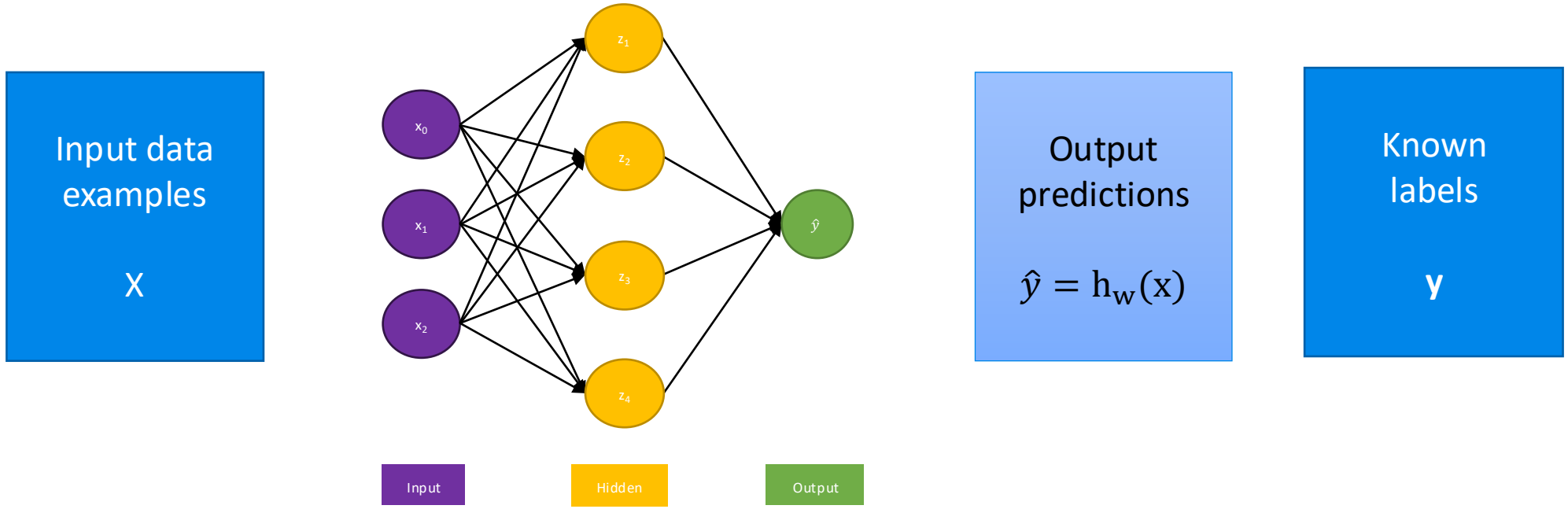
A new example
Location: [3, 3]

A toy example: Am I happy?



How close was I?

- Common approach: binary cross entropy loss
  - Used when known labels are binary (only 2 classes)



Cross Entropy

- Remember entropy?
  - Entropy = measure of uncertainty (higher entropy, more uncertainty)
  - $H(V) = -\sum_k P(v_k) log_2 P(v_k)$

- Related idea: cross entropy
  - It measures how well one distribution B approximates a true distribution A

$$H(A, B) = -\sum_k P(a_k)\, log_2 P(b_k)$$

k: possible outcomes (classes) of the random variable

  - In supervised learning:
    - A represents the **true** (data-generating) distribution
    - B is the model's **predicted** distribution

Binary Cross Entropy Loss

Notation:

- $\boldsymbol{X}$: the input matrix
- $\boldsymbol{z_i}$: the hidden units
- $\boldsymbol{s_i}$: the output layer, pre-nonlinearity
- $\boldsymbol{W}$: the weight matrix
- $\hat{\boldsymbol{y}}/\mathrm{h_w(X)}$: the estimated labels
- $\boldsymbol{y}$: the "true" labels

- We can use cross entropy as a loss
  - y – the true labels (labeled 0 or 1)
  - $h_W(x)$ - the labels predicted by the neural network

$$Loss(W) = -\frac{1}{n}\sum_{i=1}^{n} \underline{y^{(i)} \log\left(h_W\left(x^{(i)}\right)\right)} + \left(\underline{1 - y^{(i)}}\right)\log\left(\underline{1 - h_W\left(x^{(i)}\right)}\right)$$

Cross entropy for the true labels with class 1

Cross entropy for the true labels with class 0

How close was I?

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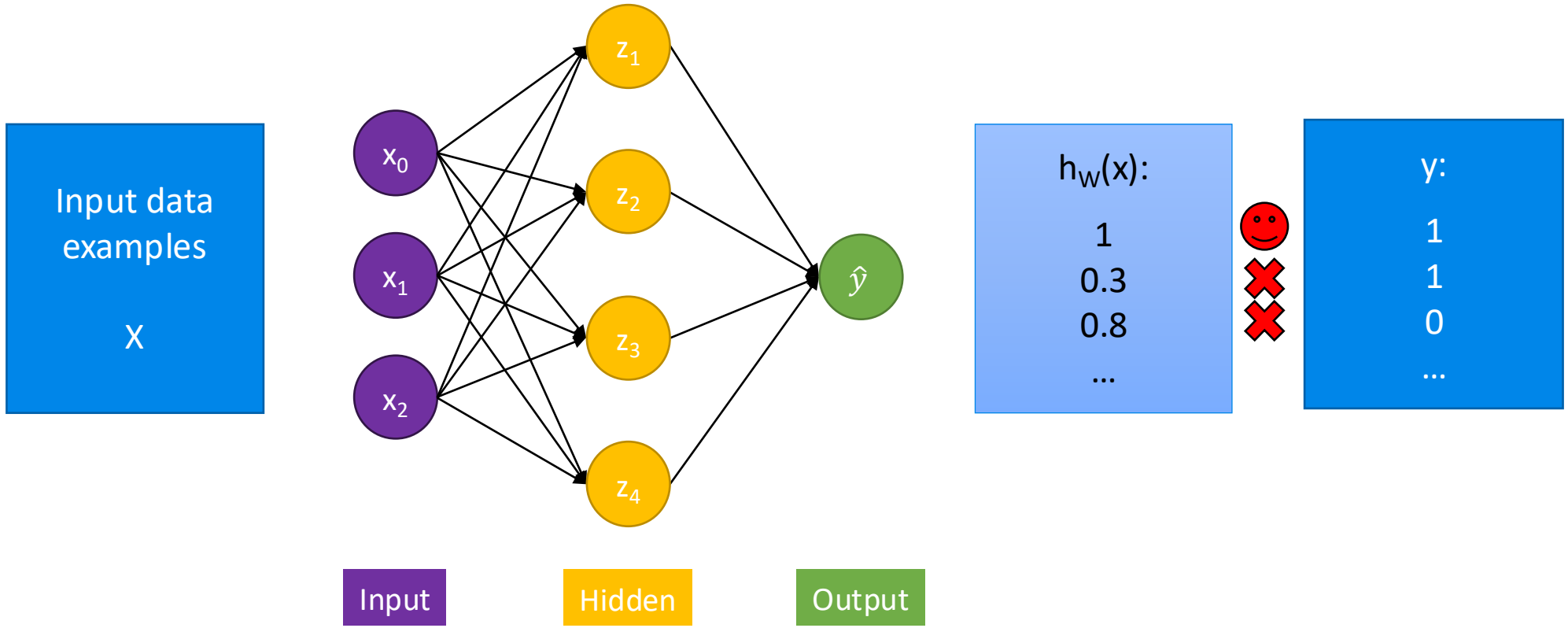

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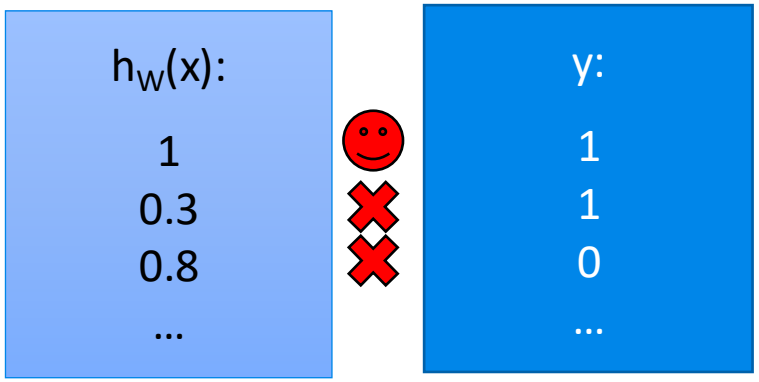

Formulate loss:

$$Loss(W) = -\frac{1}{n}\sum_i y^{(i)} \log\left(h_W\left(x^{(i)}\right)\right) + \left(1 - y^{(i)}\right)\log\left(1 - h_W\left(x^{(i)}\right)\right)$$

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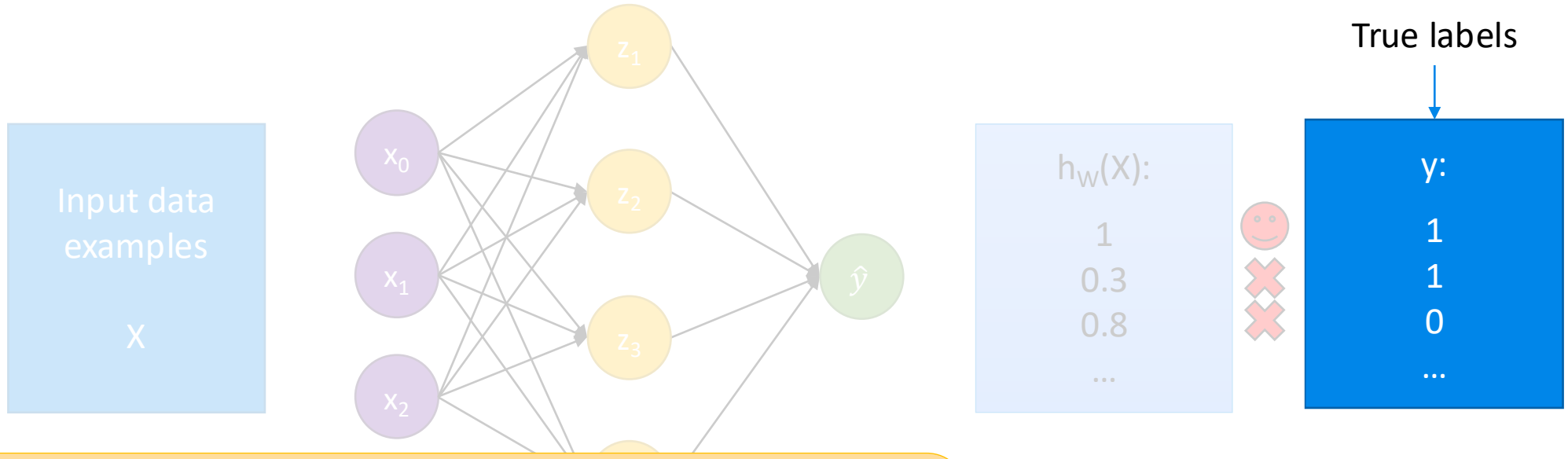

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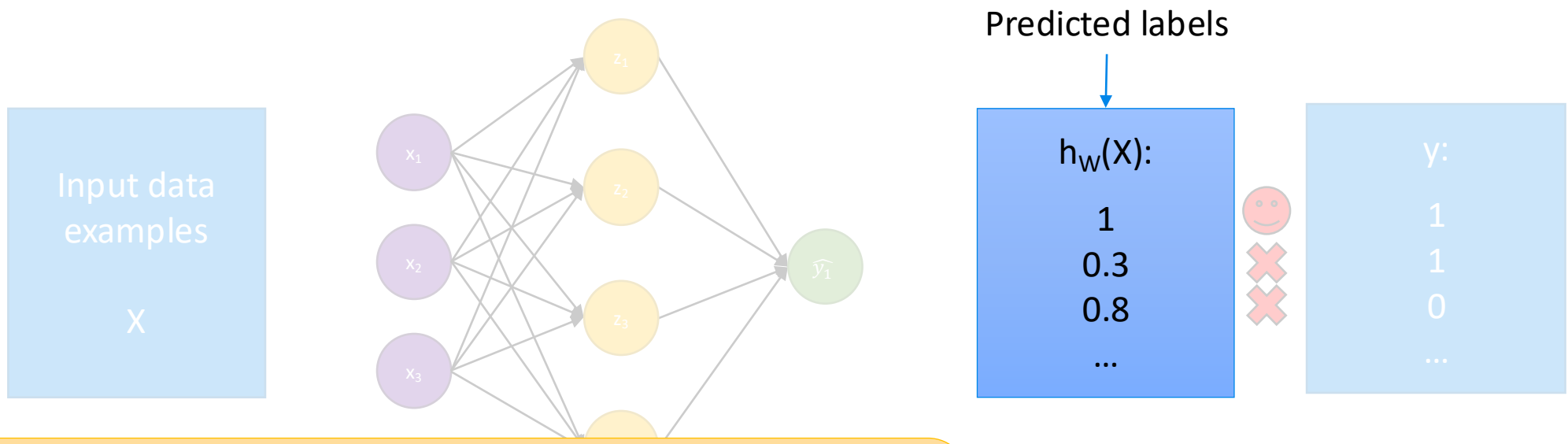

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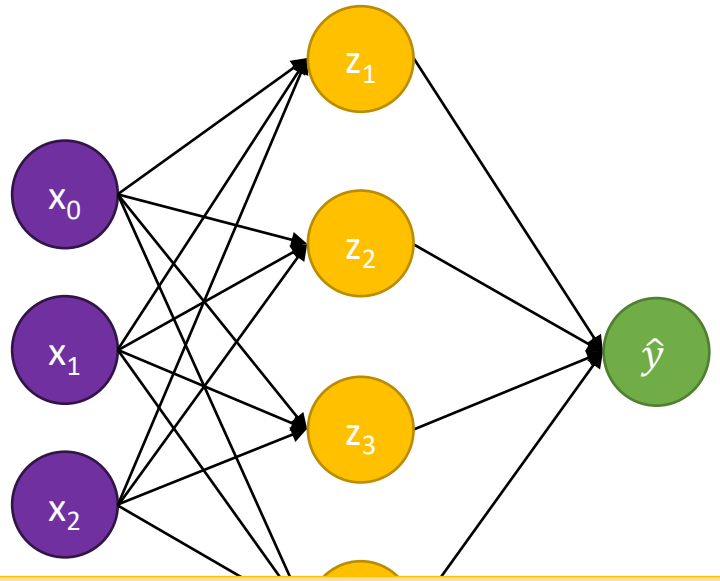
Input data examples

X



Items that are actually happy

$h_W(X)$:		y:
1	☺	1
0.3	✗	1
0.8	✗	0
…		…

Formulate loss:

$$Loss(W) = -\frac{1}{n}\sum_i \boldsymbol{y^{(i)}\, log\left(h_W\left(x^{(i)}\right)\right)} + \left(1 - y^{(i)}\right) log\left(1 - h_W\left(x^{(i)}\right)\right)$$

How close was I?

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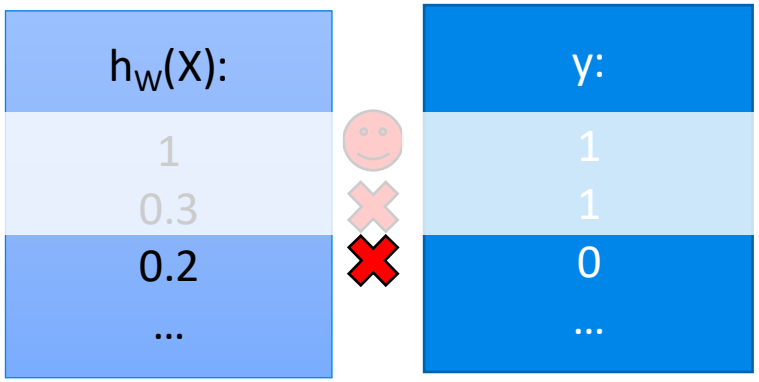
Items that are actually not happy



Formulate loss:

$$Loss(W) = -\frac{1}{n}\sum_i y^{(i)} \log\left(h_W(x^{(i)})\right) + \left(1 - y^{(i)}\right)\log\left(1 - h_W(x^{(i)})\right)$$

How close was I?

- Common approach: mean squared error (MSE) loss (how happy am I?)

Notation:

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- $\boldsymbol{W}$: the weight matrix
- $\hat{\boldsymbol{y}}/\mathrm{h_w(X)}$: the estimated labels
- $\boldsymbol{y}$: the "true" labels

Input data examples

X

$h_W(X)$:

94
32
25
…

y:

100
85
10
…

Formulate loss:

$$Loss(W) = \frac{1}{n}\sum_i \left(y^{(i)} - h_W\left(x^{(i)}\right)\right)^2$$

Now we can train the model!

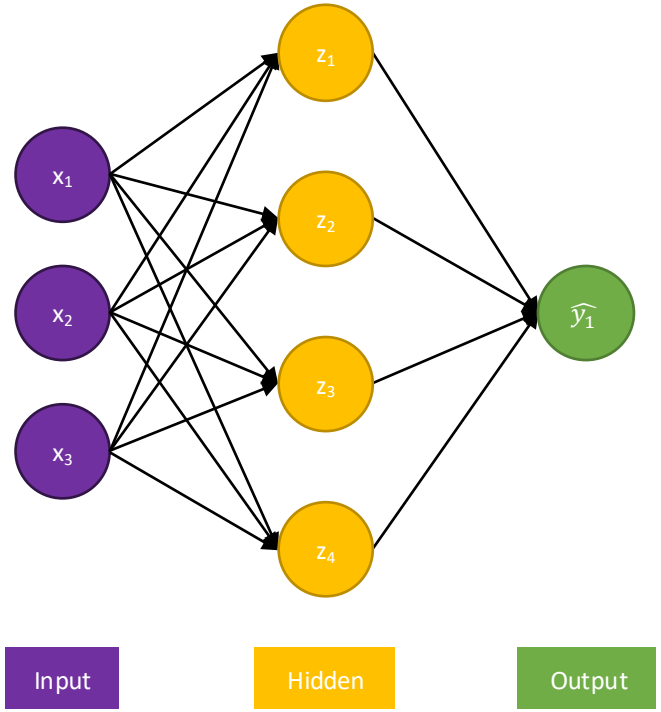
Notation:

- $\boldsymbol{X}$: the input matrix
- $\boldsymbol{z_i}$: the hidden units
- $\boldsymbol{s_i}$: the output layer, pre-nonlinearity
- $\boldsymbol{W}$: the weight matrix
- $\hat{\boldsymbol{y}}/\mathrm{h_w(x)}$: the estimated labels
- $\boldsymbol{y}$: the "true" labels

- We would like to *minimize* loss

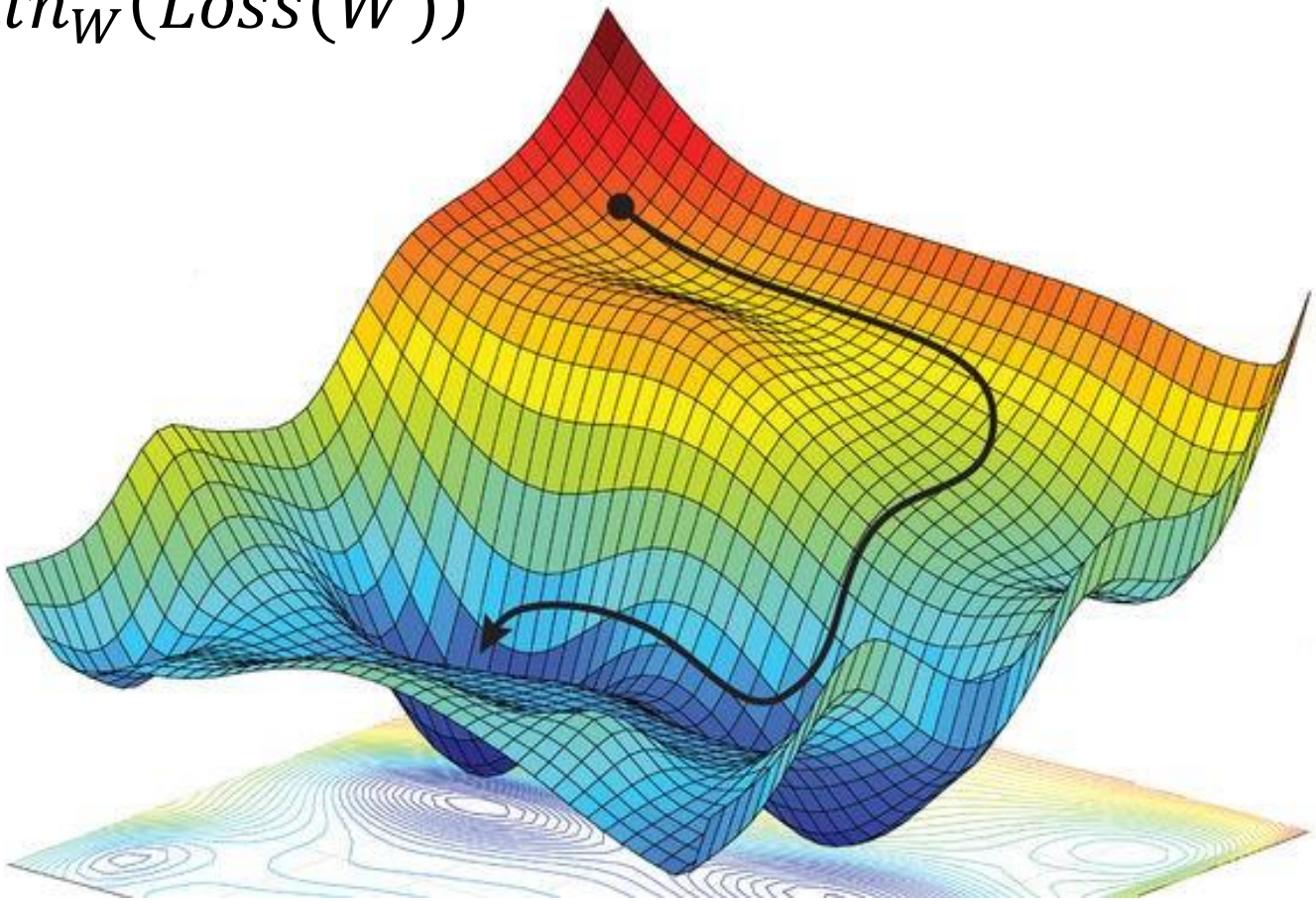
$$W^* = argmin_w \underbrace{\frac{1}{n}\sum_i \left(\mathcal{L}\left(h_W(x), y^{(i)}\right)\right)}_{Loss(W)}$$

$$W^* = argmin_W(Loss(W))$$



We need to find the right weights

$$W^* = argmin_W(Loss(W))$$



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Optimization Methods for Neural Networks

- Batch Gradient descent

- Stochastic gradient descent

Batch Gradient Descent

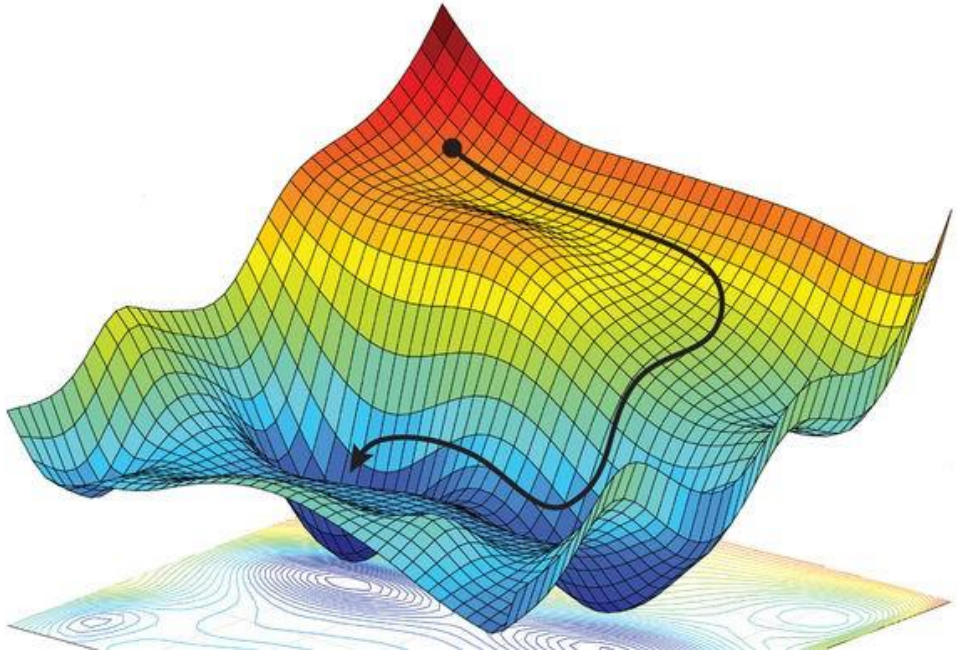
Notation:

- W: the weight matrix
- α: the learning rate

- Initialize weights randomly
- Loop until convergence:
  - Compute gradient: $\frac{\partial Loss(W)}{\partial W}$
  - Update weights: $W \leftarrow W - \alpha \frac{\partial Loss(W)}{\partial W}$
- Return weights

Learning rate

Move "downhill"



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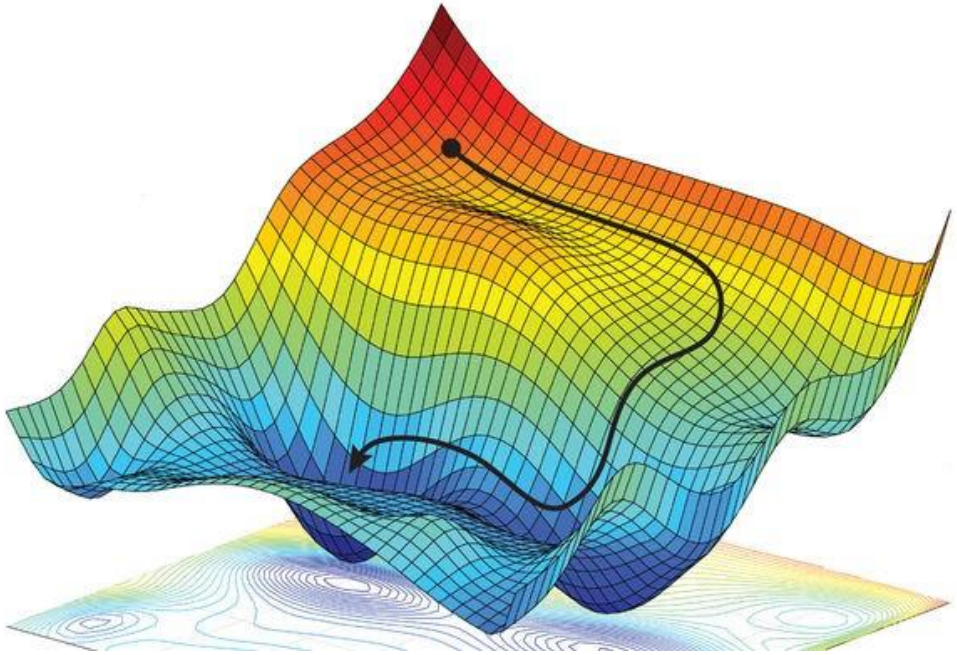
What happens if the learning rate is too small? Too big?



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Changing the learning rate

*Note: it is not necessary that the learning rate be fixed!

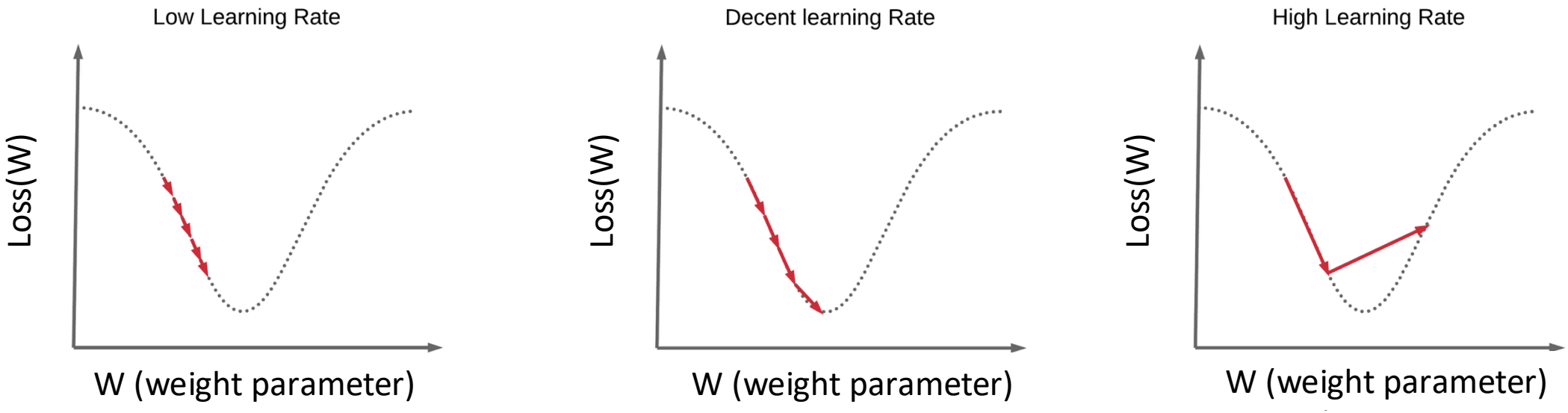


Image source: https://www.deeplearningwizard.com/deep_learning/boosting_models_pytorch/lr_scheduling/

Stochastic Gradient Descent (SGD)

- Batch gradient descent is expensive when N is large!
- Instead, SGD would update for every single example
- More efficient: approximate sum using a minibatch of examples
  - 32 / 64 / 128 common
  - Update weights with minibatch gradient

Minibatch Gradient Descent

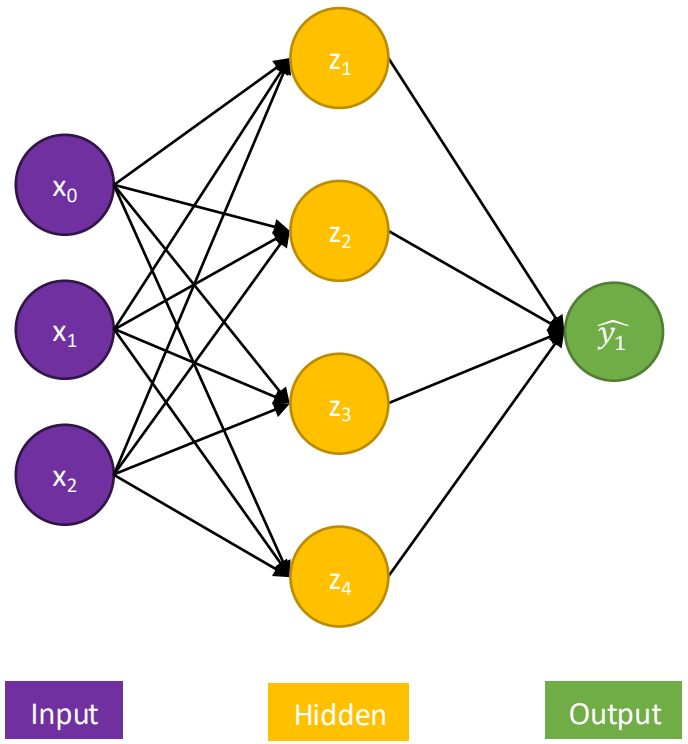
Initialize weights randomly
Loop until convergence:
    Shuffle dataset
    Loop over minibatch:
        Compute gradient for minibatch $\frac{\partial Loss\,(W)}{\partial W}$
        Update weights: $W \leftarrow W - \alpha \frac{\partial Loss(W)}{\partial W}$
Return weights

Backpropagation

Backpropagation

- Basic idea:
  - Compute effect of every weight on output
  - Work backwards from output to input

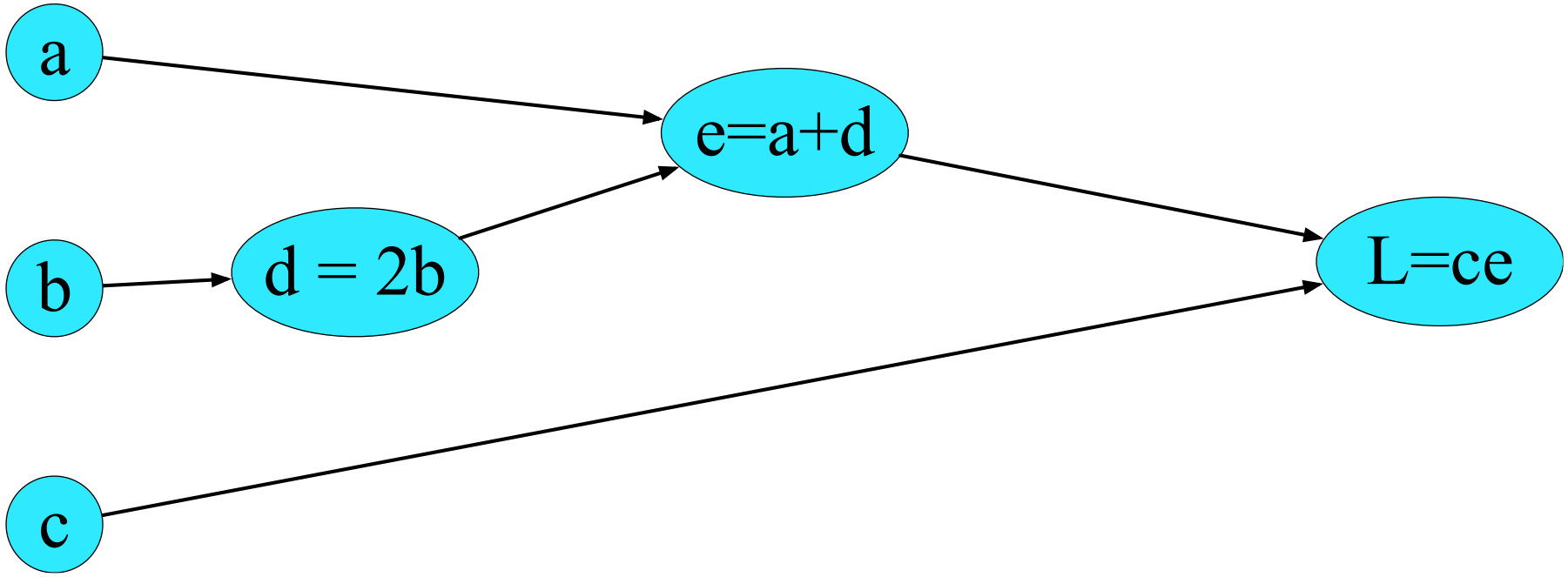
- If output is the wrong value – move weights in direction of gradient



Example of a computational graph

$$L(a,b,c) = c(a+2b)$$

$$\begin{aligned} d &= 2*b \\ e &= a+d \\ L &= c*e \end{aligned}$$



Backwards Differentiation

$$L(a,b,c) = c(a+2b)$$
$$d = 2*b$$
$$e = a+d$$
$$L = c*e$$

- Backward pass - we compute the derivatives that we'll need for the weight update

- It will help us obtain $\frac{\partial L}{\partial a}$, $\frac{\partial L}{\partial b}$, and $\frac{\partial L}{\partial c}$ to update the parameters via gradient descent

- Recall: the derivative $\frac{\partial L}{\partial a}$ tells us how much a small change in *a* affects *L*

Recall: The Chain Rule

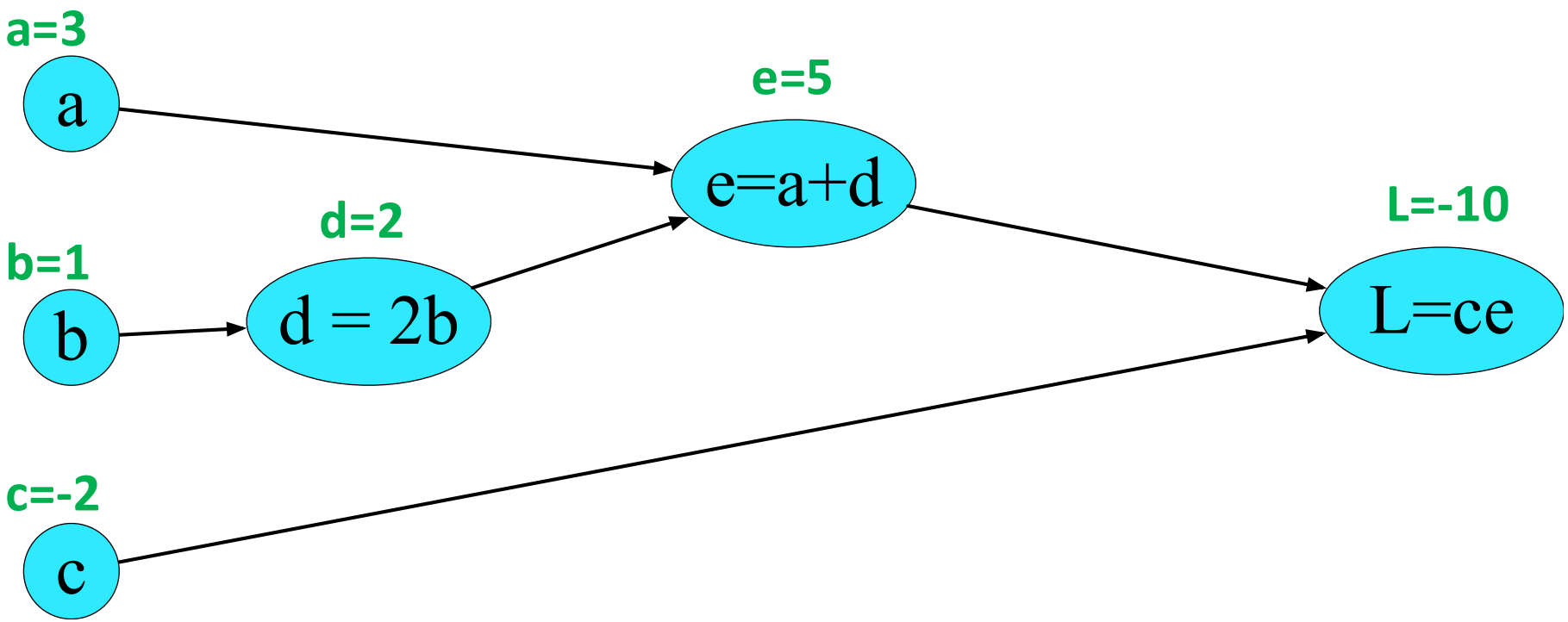
- Computing the derivative of a composite function:

- $f(x) = u(v(x))$ $$\frac{df}{dx} = \frac{du}{dv} \cdot \frac{dv}{dx}$$

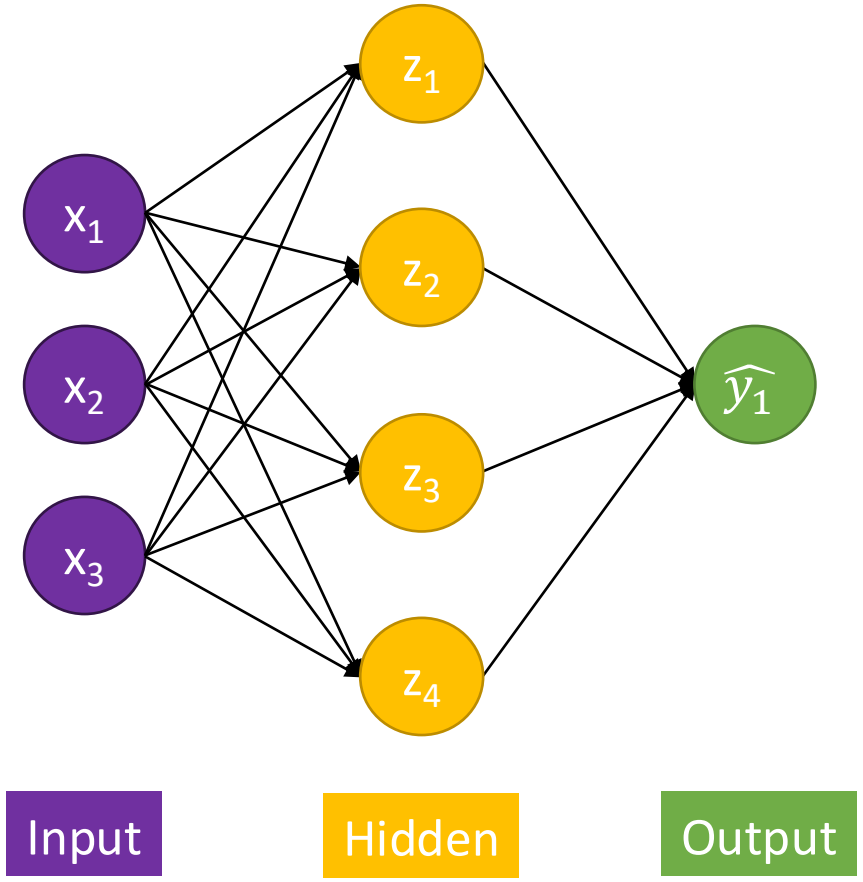
- $f(x) = u(v(w(x)))$ $$\frac{df}{dx} = \frac{du}{dv} \cdot \frac{dv}{dw} \cdot \frac{dw}{dx}$$

When poll is active respond at **PollEv.com/492**

Calculate $\frac{\partial L}{\partial a}$ using the computational graph

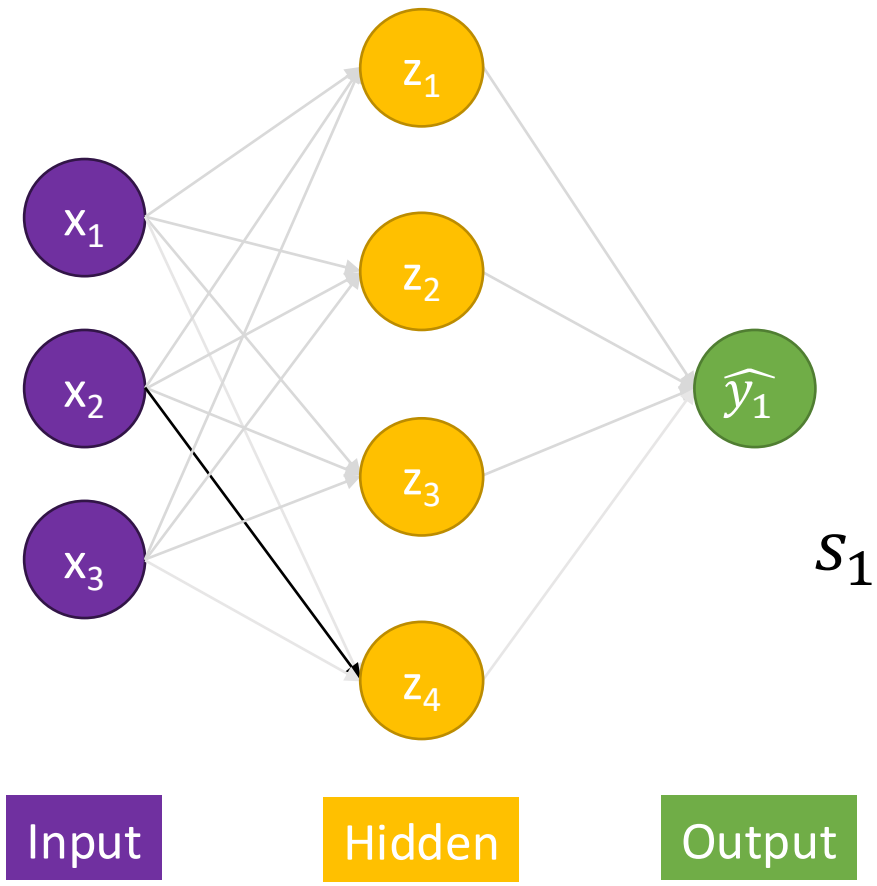


Back to training our 2-layer network



Let's step through it with focus on $w_{2,4}^{(1)}$

Back to training our 2-layer network

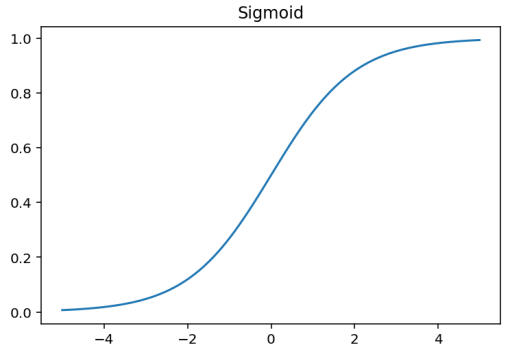


First, let's write down the forward pass equations containing $w_{2,4}^{(1)}$

$$z_4 = g\left(w_{0,4}^{(1)} + x_1 w_{1,4}^{(1)} + x_2 w_{2,4}^{(1)} + x_3 w_{3,4}^{(1)}\right)$$

$$s_1 = w_{0,1}^{(2)} + z_1 w_{1,1}^{(2)} + z_2 w_{2,1}^{(2)} + z_3 w_{3,1}^{(2)} + z_4 w_{4,1}^{(2)}$$

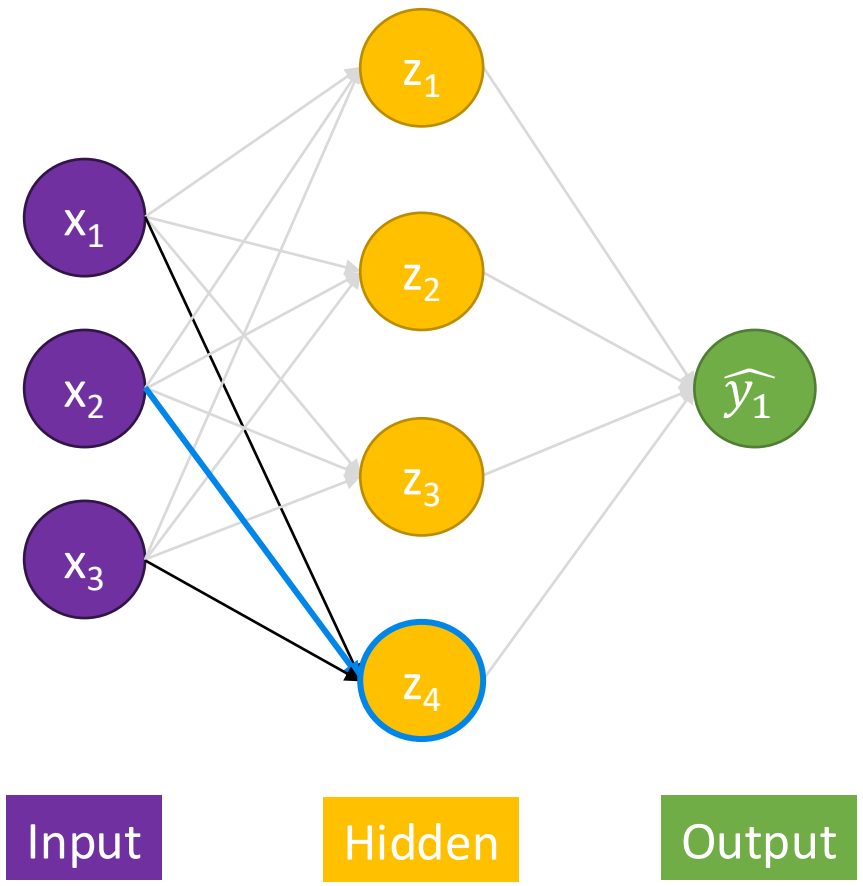
$$\hat{y}_1 = \sigma(s_1)$$



where $\sigma(s) = \frac{1}{1+e^{-s}}$

and g is ReLU

Back to training our 2-layer network

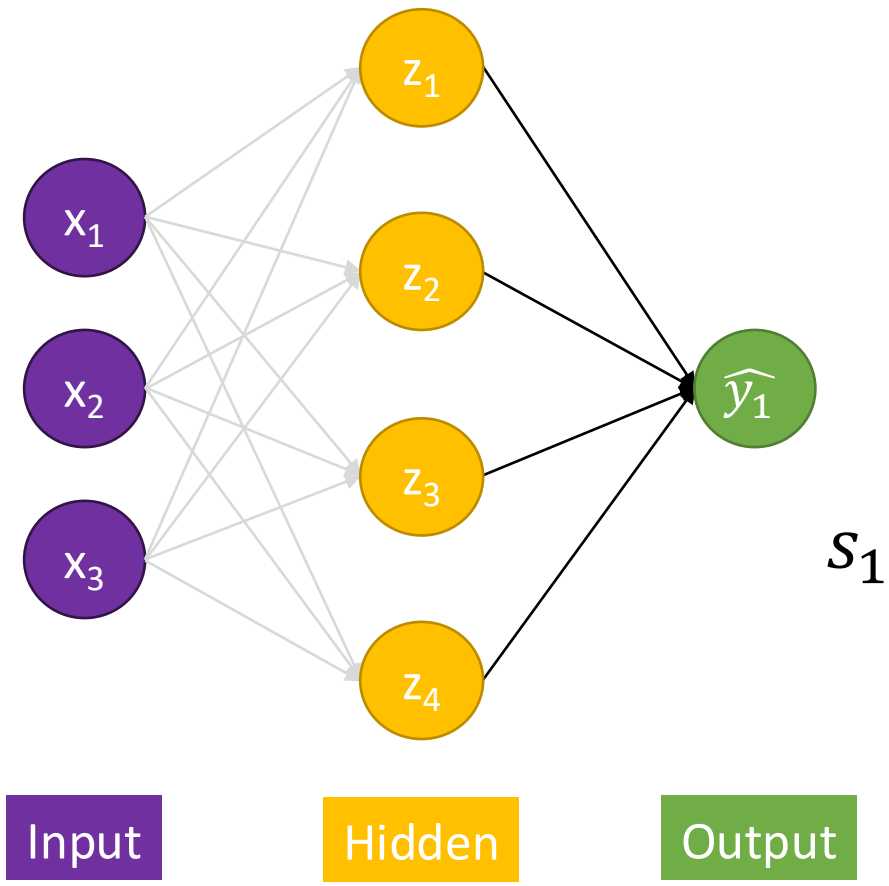


Let's step through it step by step with focus on $w_{2,4}^{(1)}$

$$z_4 = g\left(w_{0,4}^{(1)} + x_1 w_{1,4}^{(1)} + x_2 w_{2,4}^{(1)} + x_3 w_{3,4}^{(1)}\right)$$

where g is ReLU

Back to training our 2-layer network



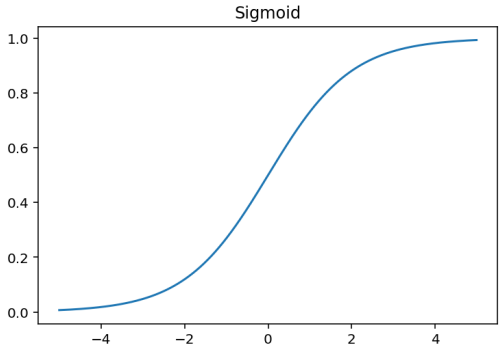
Let's step through it step by step with focus on z_4

Now with the output layer:

$$s_1 = w_{0,1}^{(2)} + z_1 w_{1,1}^{(2)} + z_2 w_{2,1}^{(2)} + z_3 w_{3,1}^{(2)} + z_4 w_{4,1}^{(2)}$$

$$\hat{y}_1 = \sigma(s_1)$$

where $\sigma(s) = \frac{1}{1+e^{-s}}$



Recall: Binary Cross Entropy Loss

- A measure for how far off the current answer is to the right answer

$$Loss(y, \hat{y}) = -[y \ \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

- Derivative:

$$\frac{\partial Loss}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{(1 - y)}{(1 - \hat{y})}$$

Updating $w_{2,4}^{(1)}$

$$Loss = -[y \log \hat{y}_1 + (1-y)\log(1-\hat{y}_1)]$$
$$\hat{y}_1 = \sigma(s_1)$$
$$s_1 = w_{0,1}^{(2)} + z_1 w_{1,1}^{(2)} + z_2 w_{2,1}^{(2)} + z_3 w_{3,1}^{(2)} + z_4 w_{4,1}^{(2)}$$
$$z_4 = g\left(w_{0,4}^{(1)} + x_1 w_{1,4}^{(1)} + x_2 w_{2,4}^{(1)} + x_3 w_{3,4}^{(1)}\right)$$

- We will need $\frac{\partial Loss}{\partial w_{2,4}^{(1)}}$
- Chain rule:

$$\frac{\partial Loss}{\partial w_{2,4}^{(1)}} = \frac{\partial Loss}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial s_1} \cdot \frac{\partial s_1}{\partial z_4} \cdot \frac{\partial z_4}{\partial w_{2,4}^{(1)}}$$

Derivative of the Loss

Derivative of the activation (sigmoid)

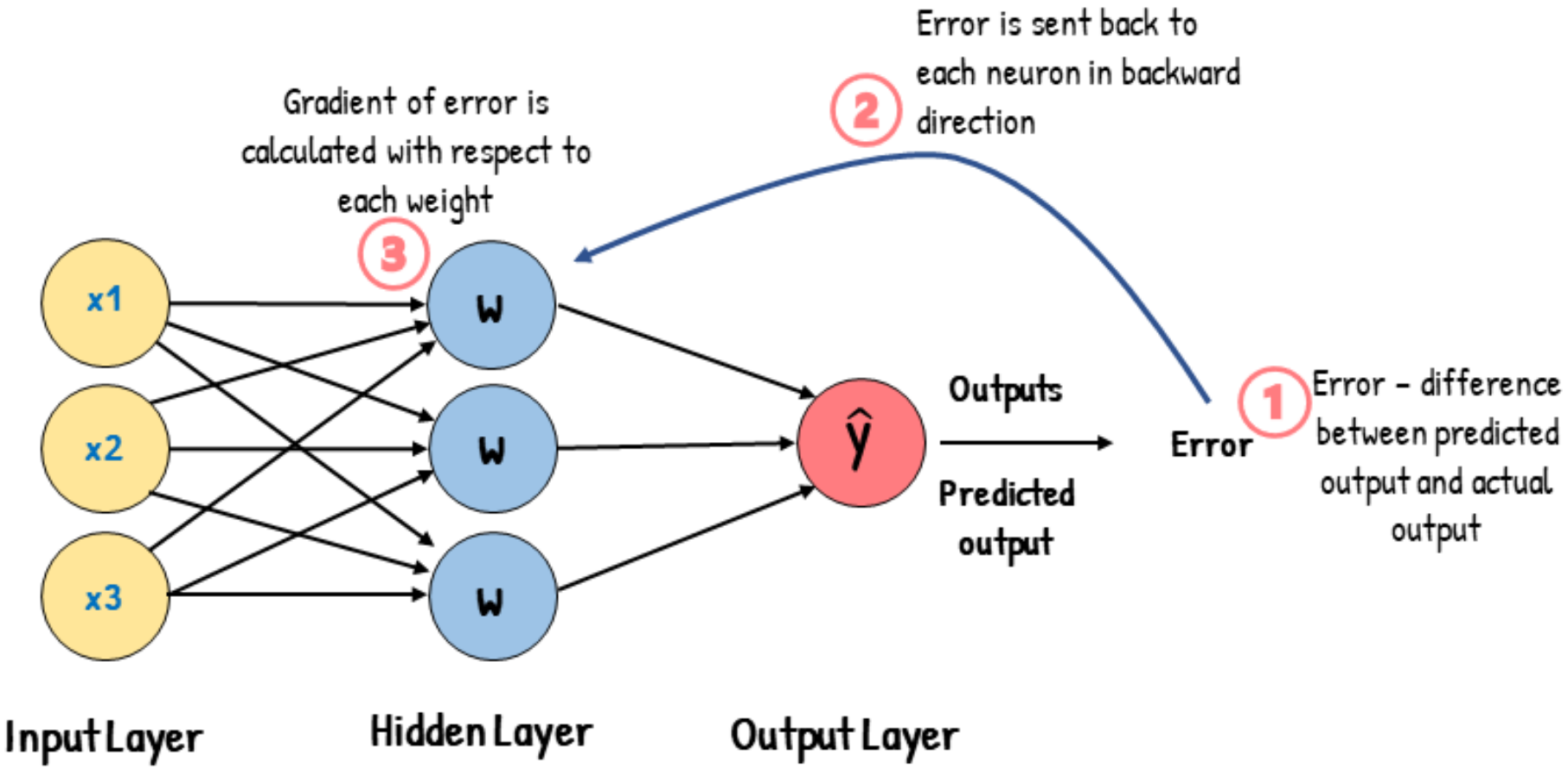
Derivative of the weighted sum

Derivative of activation (ReLU) and weighted sum

- Then, update rule: $w_{2,4}^{(1)} \leftarrow w_{2,4}^{(1)} - \alpha \frac{\partial Loss}{\partial w_{2,4}^{(1)}}$

We will do the same for every weight in the neural network

Summary: Backprop Computes Gradients needed for Gradient Descent Algorithm



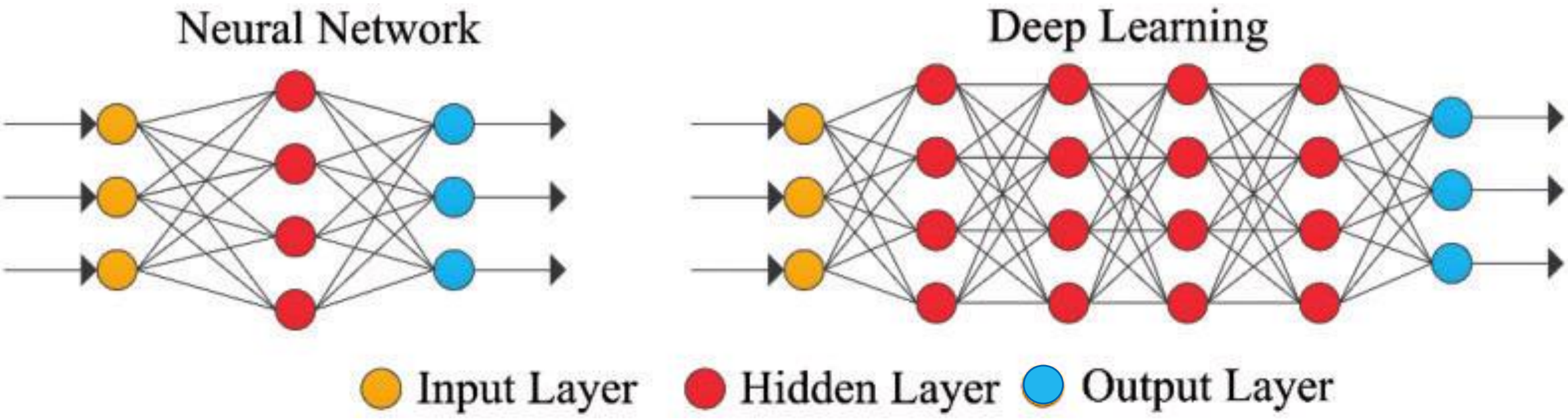
Source: https://www.analyticsvidhya.com/blog/2023/01/gradient-descent-vs-backpropagation-whats-the-difference/

Notes on Gradient Descent

- No convergence guarantees due to many local minima
- Problem of finding consistent weights is NP-complete
- Performance dependent on network structure
  - Need sufficient hidden nodes to express target
  - But too many nodes leads to overfitting + slow training!

Deep Learning: A Little Bit of History

From Neural Networks to Deep Learning

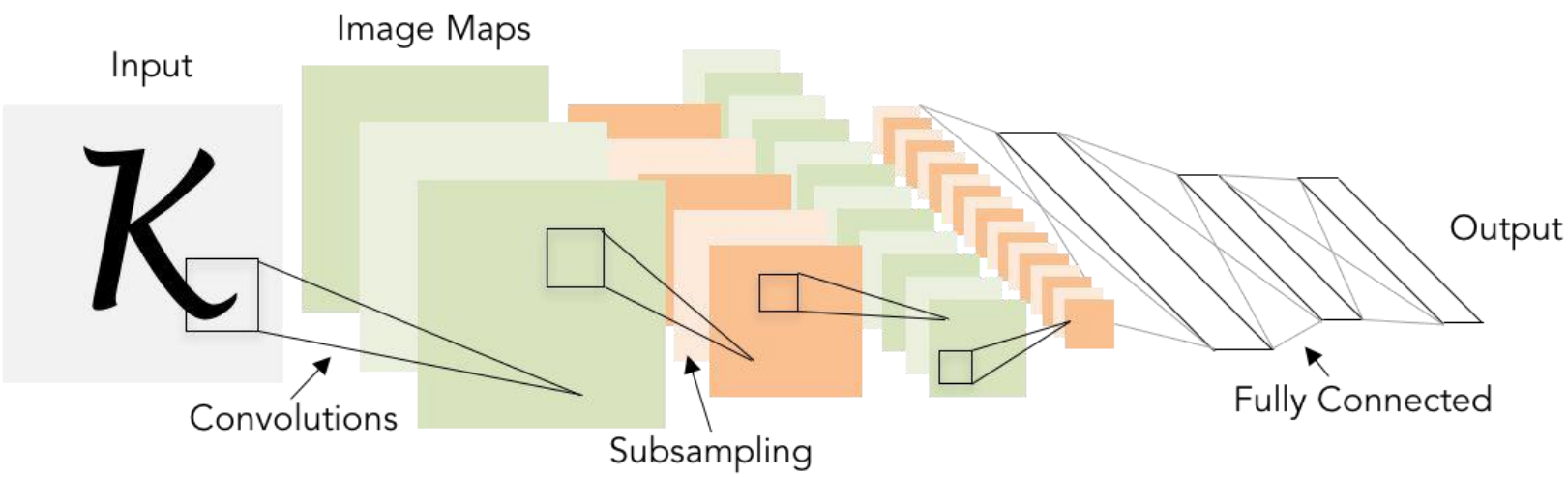


Source: Wanli Xing

A bit of history:

Gradient-based learning applied to document recognition

[LeCun, Bottou, Bengio, Haffner 1998]

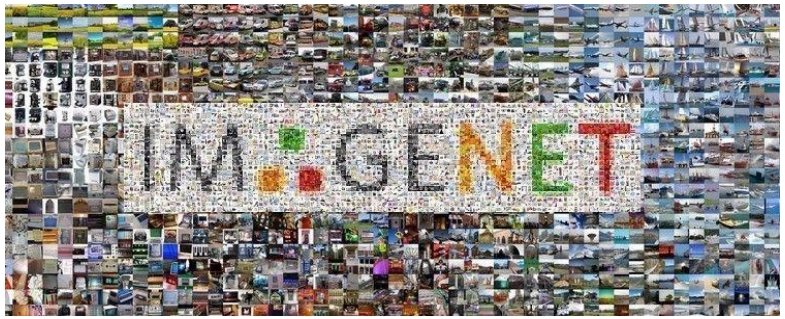


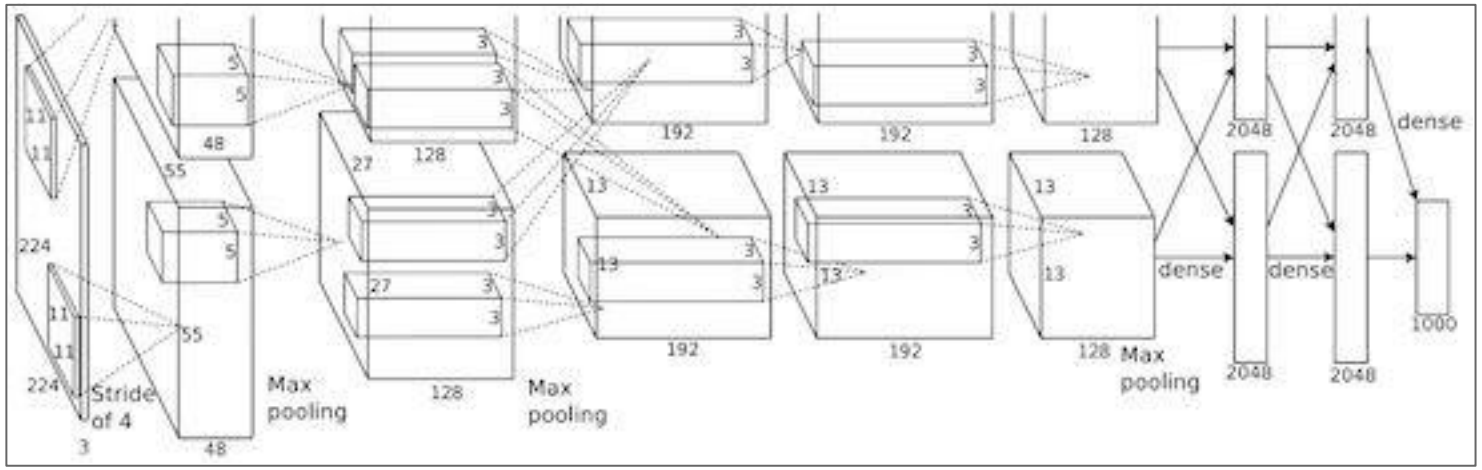
LeNet-5

LeNet-1 from 1993

A bit of history:

ImageNet Classification with Deep Convolutional Neural Networks *[Krizhevsky, Sutskever, Hinton, 2012]*





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"AlexNet"

AlexNet Results

Classification

Retrieval

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Convolutional Neural Nets (CNNs)

CNNs

original: Fukushima, Kunihiko. "Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position." *Biological cybernetics* 36.4 (1980): 193-202.

breakthrough: Yann Lecun, Léon Bottou, Yoshua Bengio, Patrick Haffner. Gradient-based learning applied to document recognition. Proceedings of the IEEE, 1998, 86 (11), pp.2278-2324. ff10.1109/5.726791ff. ffhal03926082f

Able to recognize spatial patterns in grid-like data.

Applications in computer vision

- Image classification
- Autonomous vehicles
- Facial recognition
- …

But is sometimes applied to other tasks as well (time series data: sequence of vectors)

- Sentence classification
- Speech recognition
- …

Image Processing: Problems w/ FCNN

RGB image: three integers (0-255) per pixel

Red Green Blue

- (0, 255, 0) is green
- (144, 83, 255) is lighter purple
- (60, 0, 168) is darker purple

1000x1000 RGB image:
1000x1000x3 = 3M integers

FCNN to input 1000x1000 RGB image, with 1000 hidden units
So $W^{[1]}$ would have shape (1000, 3M), so 3B weights!

- Great danger of overfitting
- Massive computational requirements
- The units in a FC layer don't know their spatial relationships.
- An image that's shifted a little bit appears completely different!

Design Principles of CNNs

A Dog

Spatial structure

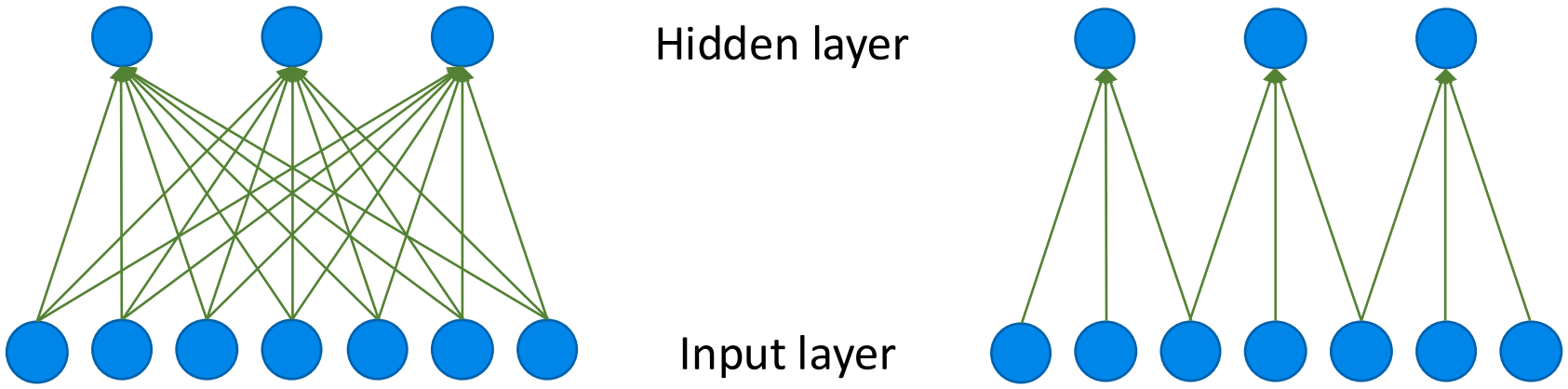
Pixels of an entity should be adequately distributed in space

Still a Dog

Spatial invariance

The semantics of an entity should not be affected much by its position

CNNs Property 1: Local Connectivity



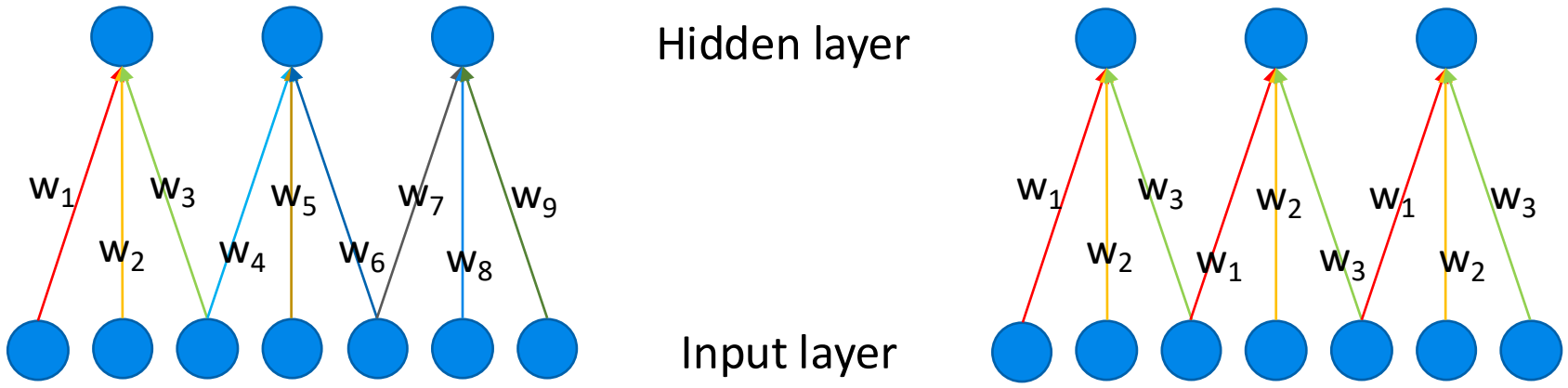
Global connectivity

Local connectivity

- # input units (neurons): 7
- # hidden units: 3
- Number of parameters
  - Global connectivity: 3 x 7 = 21
  - Local connectivity: 3 x 3 = 9

Spatial structure helps us reduce the number of connections!

CNNs Property 2: Weight Sharing



Without weight sharing

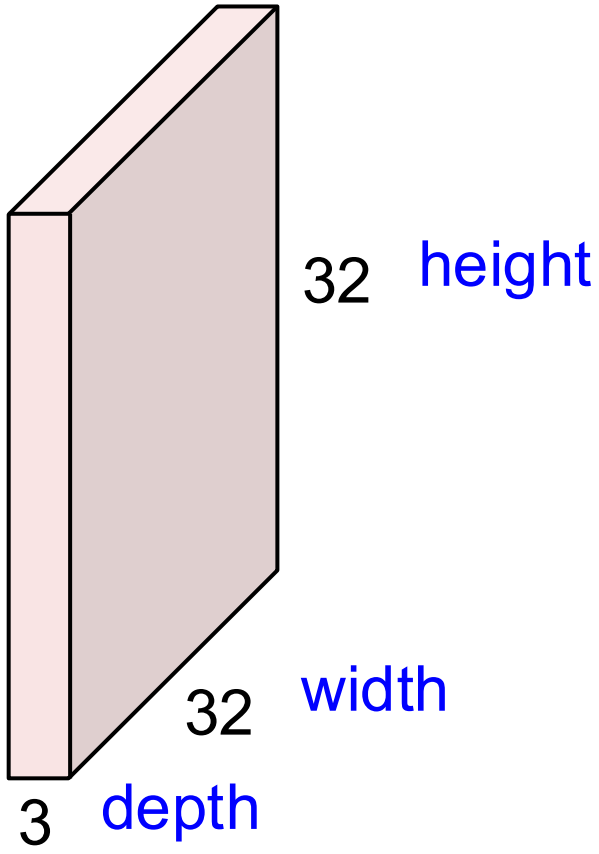
With weight sharing

- # input units (neurons): 7
- # hidden units: 3
- Number of parameters
  - Without weight sharing: 3 x 3 = 9
  - With weight sharing : 3 x 1 = 3

Spatial invariance helps us reduce number of weights (parameters to learn)!

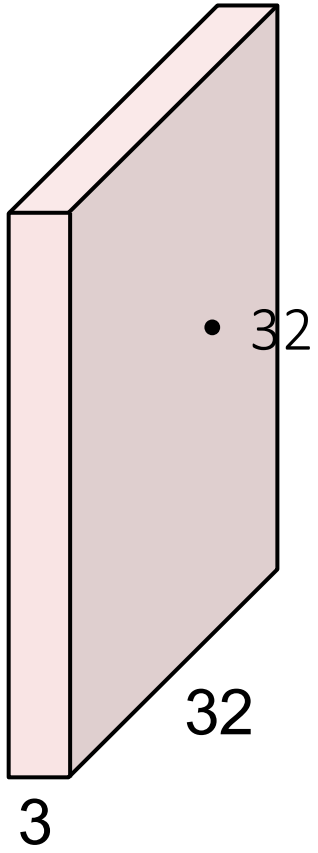
Convolution Layer

32x32x3 image -> preserve spatial structure

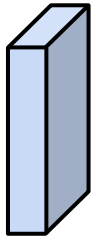


Convolution Layer

- 32x32x3 image



- 5x5x3 filter

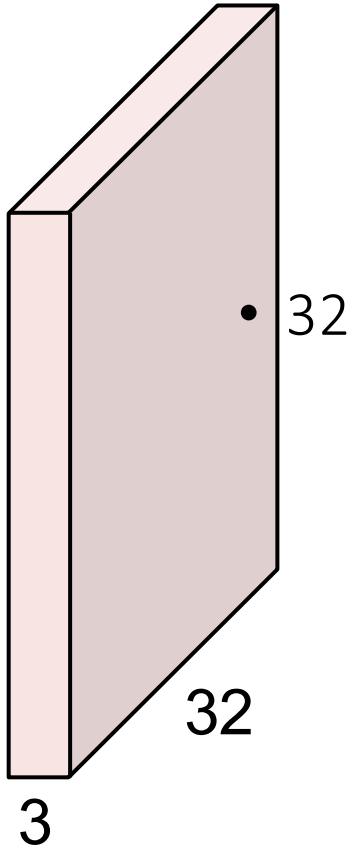


- **Convolve** the filter with the image
- i.e. “slide over the image spatially, computing dot products”

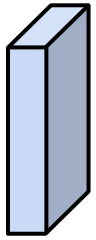
Convolution Layer

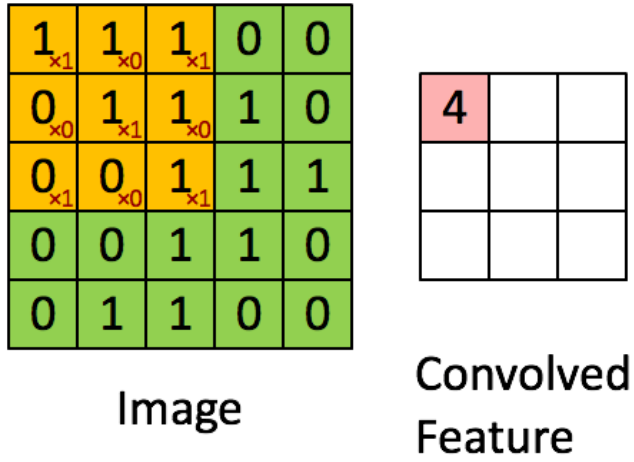
Filters always extend the full depth of the input volume

- 32x32x3 image



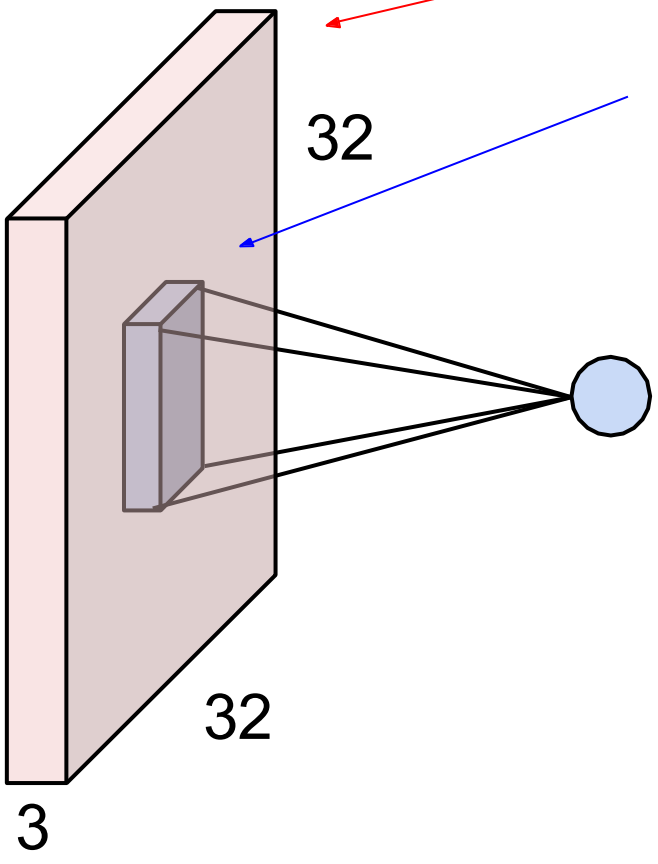
- 5x5x3 filter





- **Convolve** the filter with the image
- i.e. "slide over the image spatially, computing dot products"

Convolution Layer



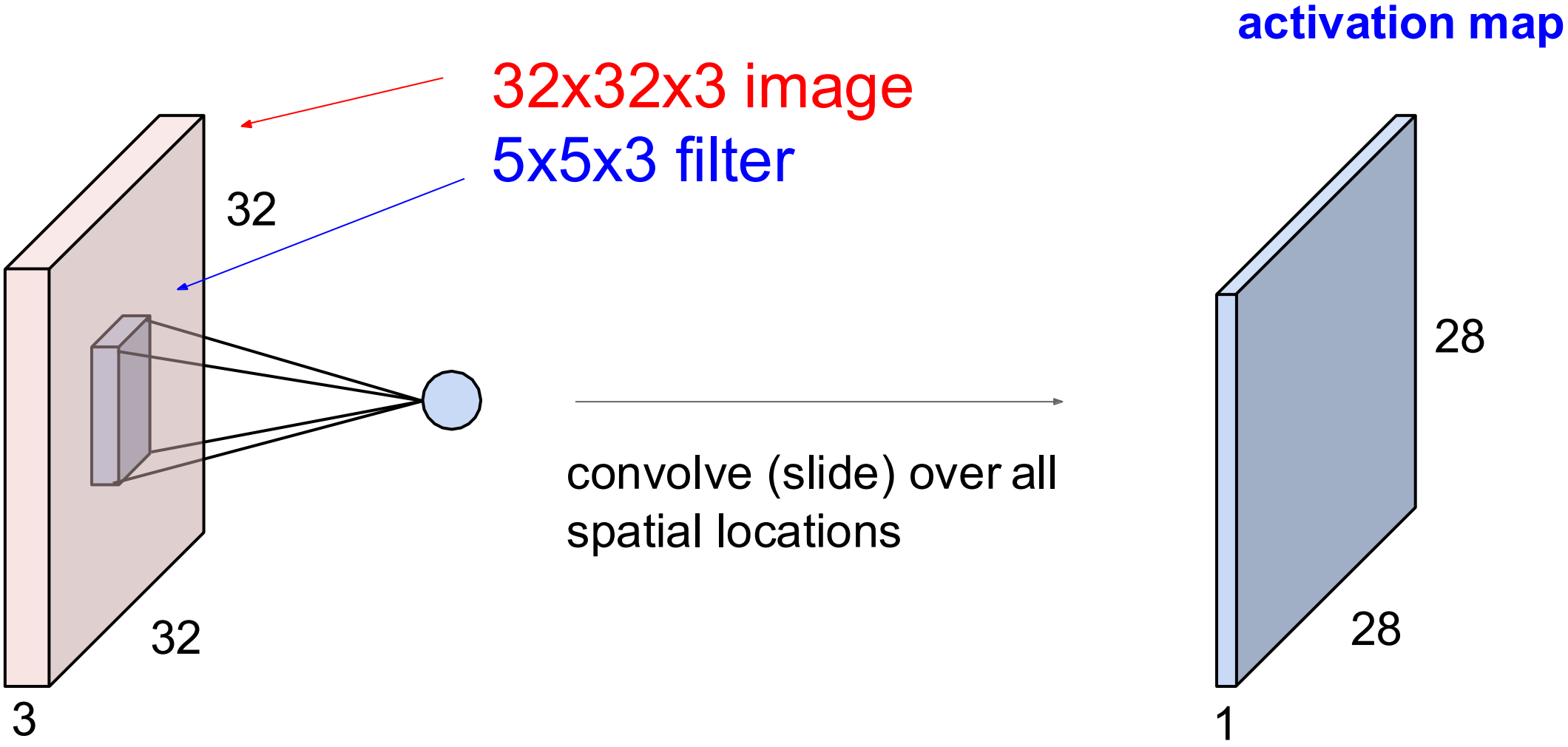
32x32x3 image

5x5x3 filter w

1 number:
the result of taking a dot product between the filter and a small 5x5x3 chunk of the image (i.e. 5*5*3 = 75-dimensional dot product + bias)

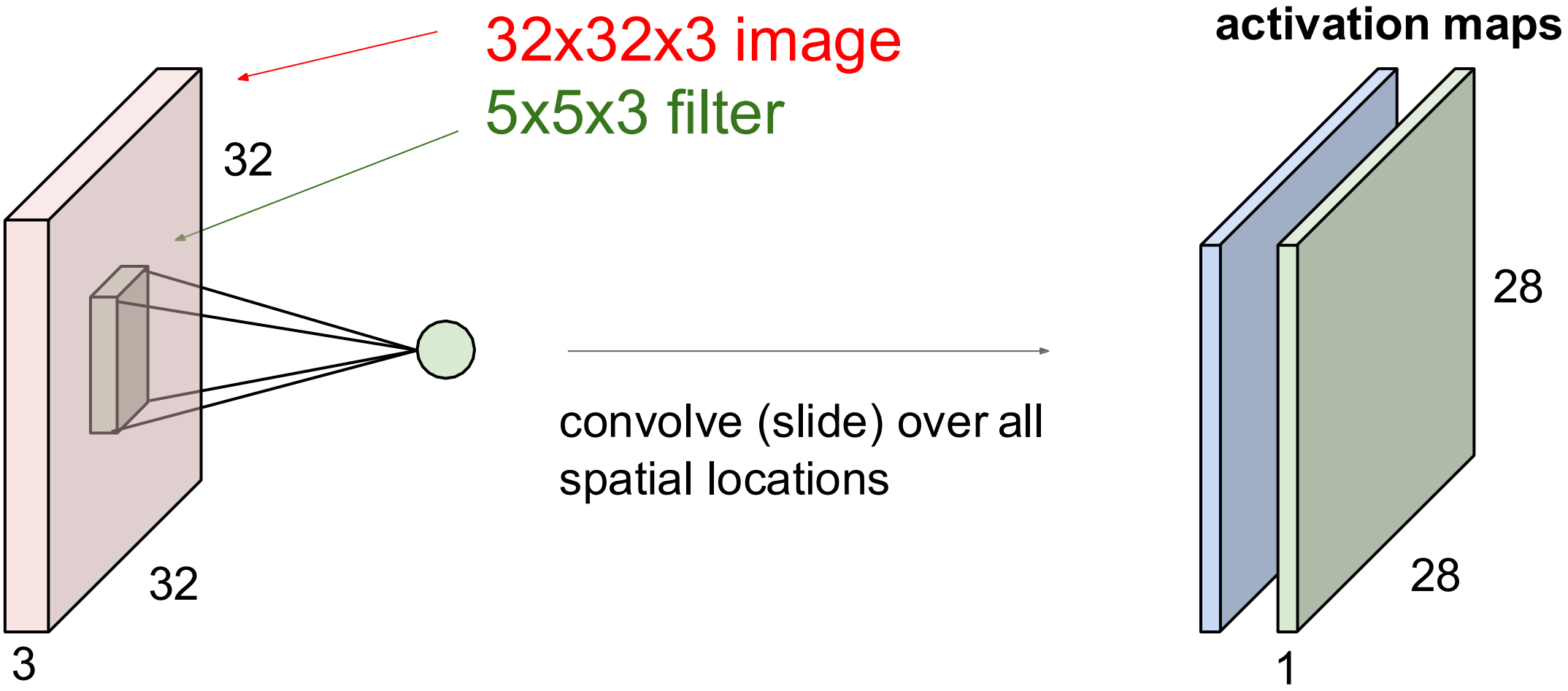
$$w^T x + b$$

Convolution Layer

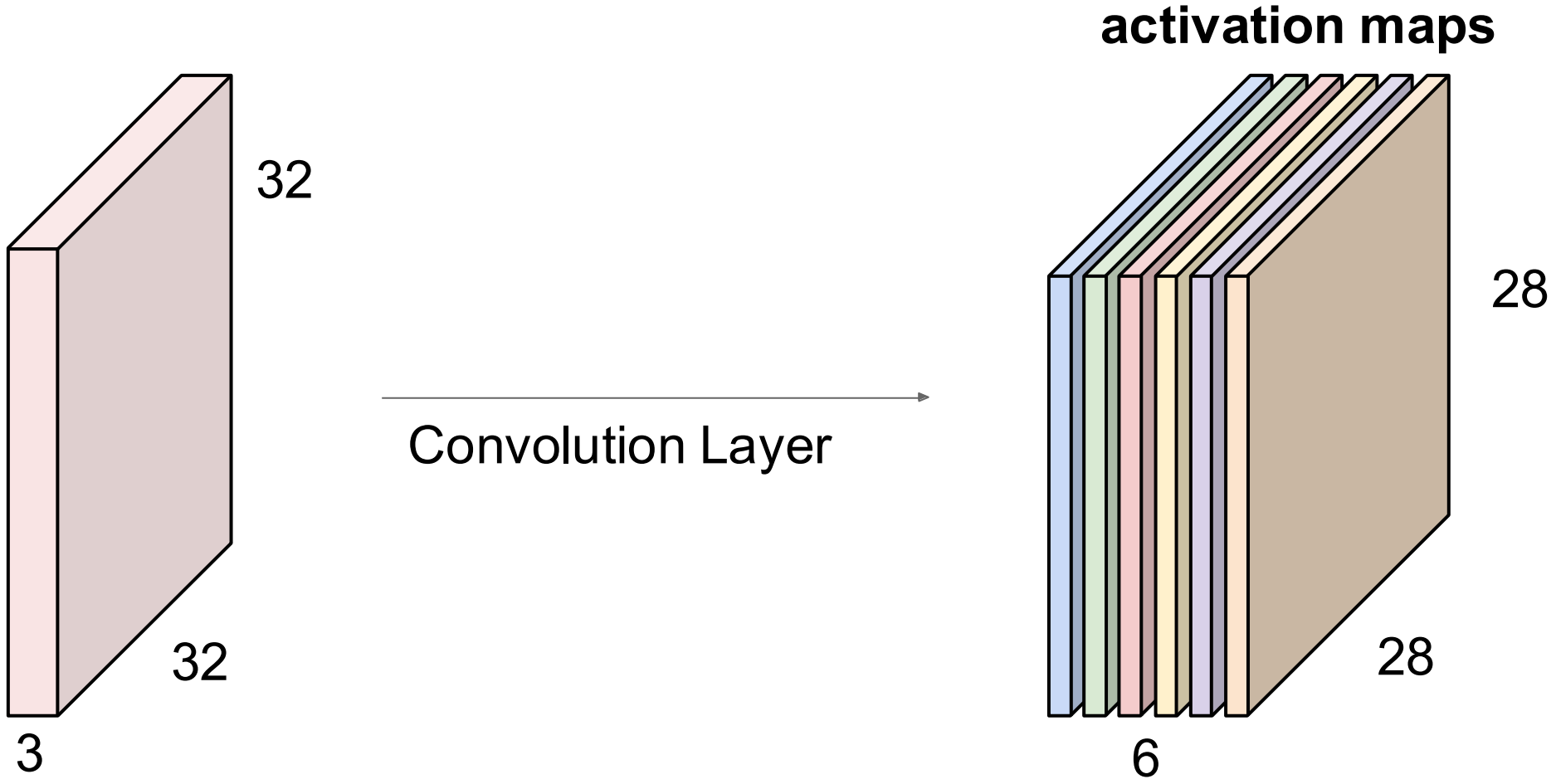


Convolution Layer

consider a second, green filter

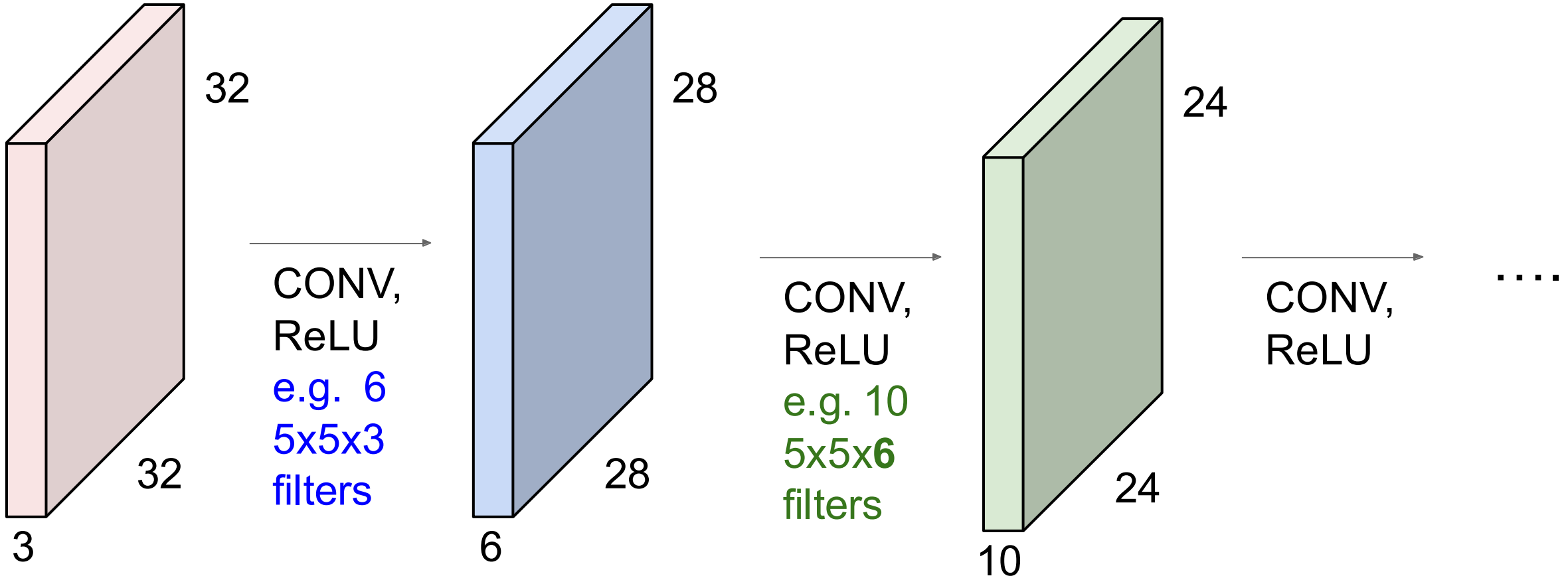


For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:



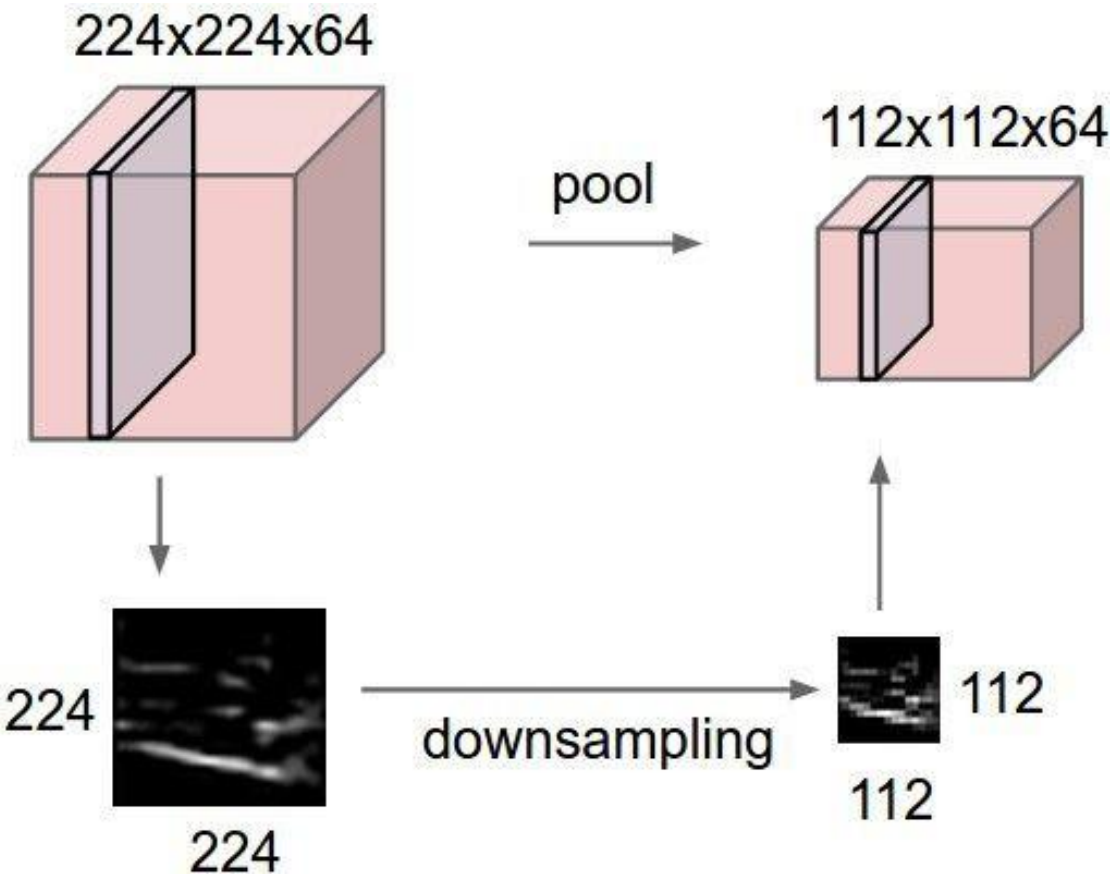
We stack these up to get a "new image" of size 28x28x6!

Preview: ConvNet is a sequence of Convolution Layers, interspersed with activation functions



Pooling layer

- makes the representations smaller and more manageable
- operates over each activation map independently:



MAX POOLING

Single depth slice

x

1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

y

max pool with 2x2 filters
and stride 2 ()

6	8
3	4

Stride: number of pixels we're stepping over as we slide our filter across the input image

Putting them together

- Local connectivity
- Weight sharing
- Handling multiple input channels
- Handling multiple output maps

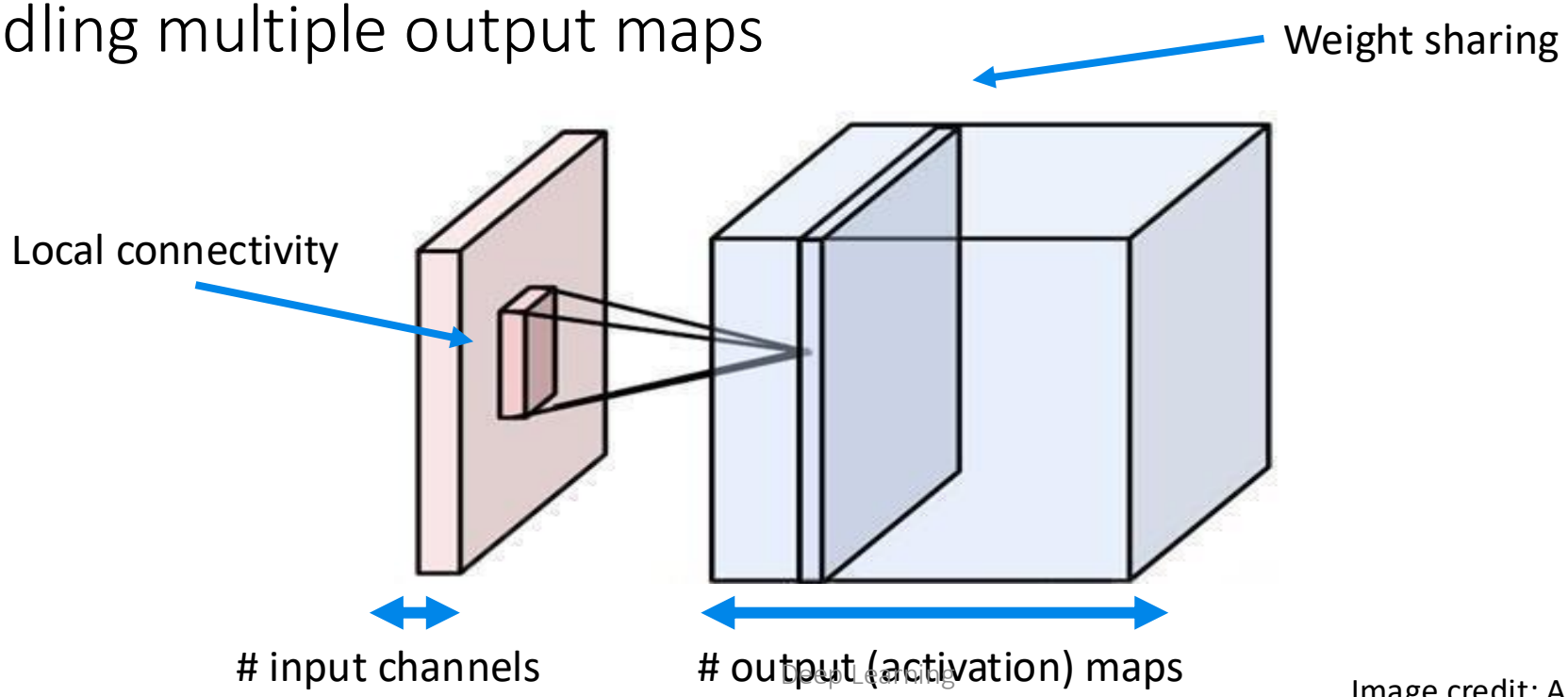
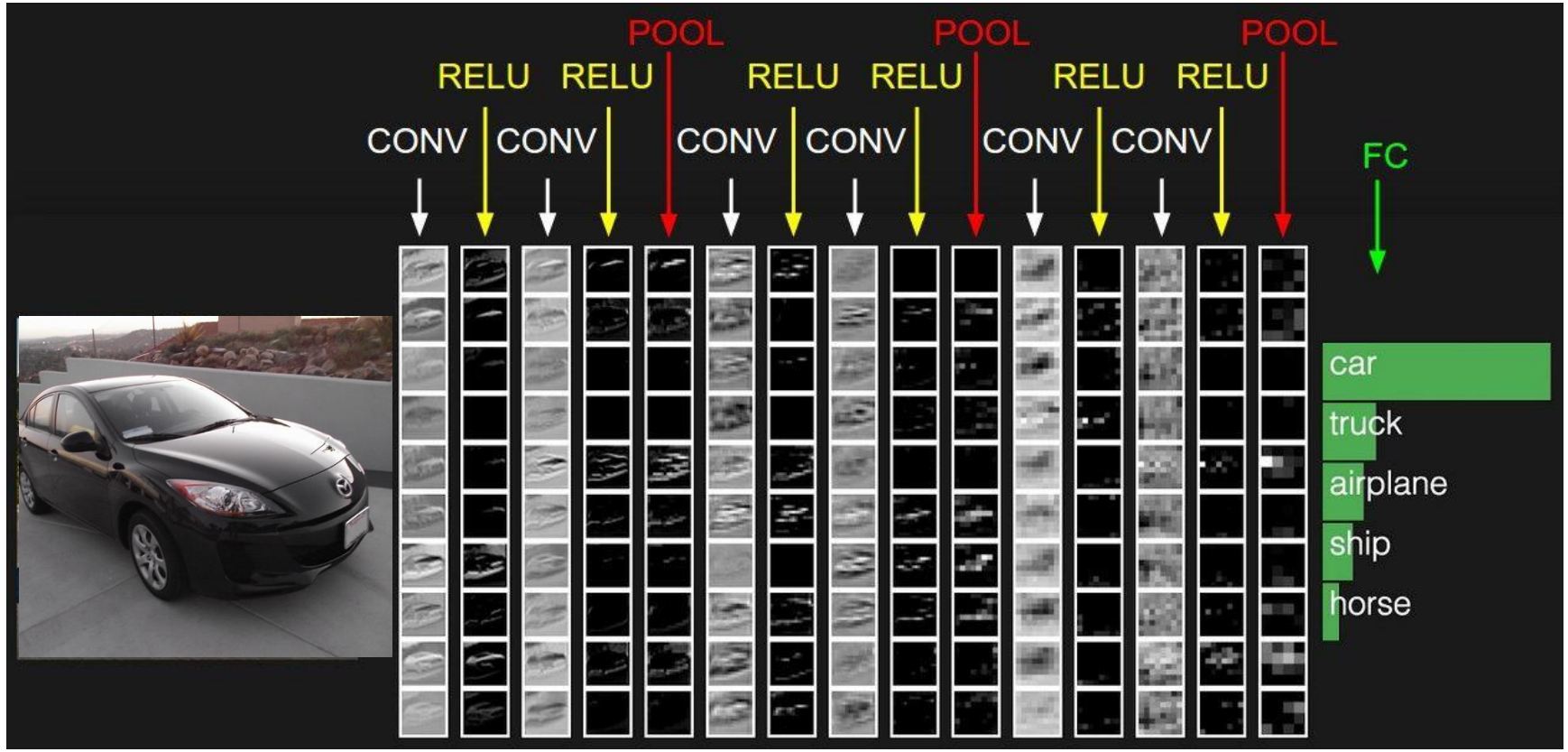


Image credit: A. Karpathy

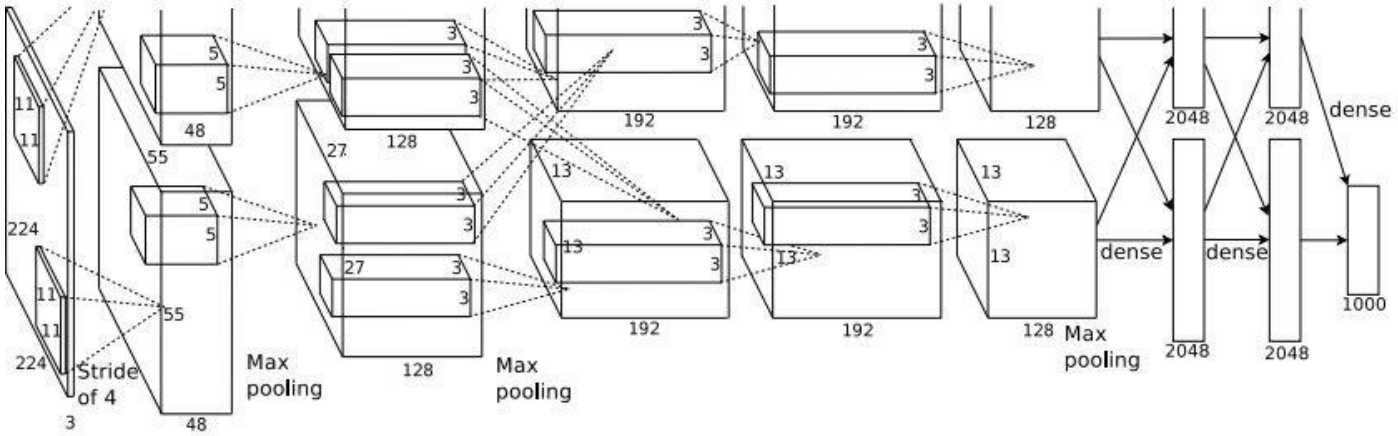
Fully Connected Layer (FC layer)

- Contains neurons that connect to the entire input volume, as in ordinary Neural Networks



Case Study: AlexNet

[Krizhevsky et al. 2012]



Full (simplified) AlexNet architecture:
[227x227x3] INPUT
[55x55x96] CONV1: 96 11x11 filters at stride 4, pad 0
[27x27x96] MAX POOL1: 3x3 filters at stride 2
[27x27x96] NORM1: Normalization layer
[27x27x256] CONV2: 256 5x5 filters at stride 1, pad 2
[13x13x256] MAX POOL2: 3x3 filters at stride 2
[13x13x256] NORM2: Normalization layer
[13x13x384] CONV3: 384 3x3 filters at stride 1, pad 1
[13x13x384] CONV4: 384 3x3 filters at stride 1, pad 1
[13x13x256] CONV5: 256 3x3 filters at stride 1, pad 1
[6x6x256] MAX POOL3: 3x3 filters at stride 2
[4096] FC6: 4096 neurons
[4096] FC7: 4096 neurons
[1000] FC8: 1000 neurons (class scores)

- First deep learning model to utilize Graphics Processing Units (GPUs) for training
- Deeper than previous neural nets
- + several other tricks, e.g., **dropout**

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